

The Limits of Merit:

Career Pipelines and Representation in US Law Enforcement

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Abstract

Civil service systems promise equal access to government employment, yet the bureaucracies they govern often remain unrepresentative: US police departments are systematically Whiter and more Republican than their jurisdictions. Scholarship and police reforms have focused on written entry exams, but bureaucratic careers extend well beyond entry. Drawing on novel data linking exam, career, and voter records for 99,000 exam passers and 58,000 NYPD officers, this study decomposes these disparities across the career pipeline. The application and written exam produce limited sorting. Instead, even among applicants with comparable exam scores, post-exam hiring favors Republican and White candidates, who are more likely to be promoted, receive awards, and remain in the agency. Equalizing post-exam hiring alone would increase appointments of Democratic and Black candidates by 78% and 27%. These findings redirect attention from the entry exam to the discretionary procedures that follow, and reveal how formally meritocratic institutions sustain unequal representation.

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1 Introduction

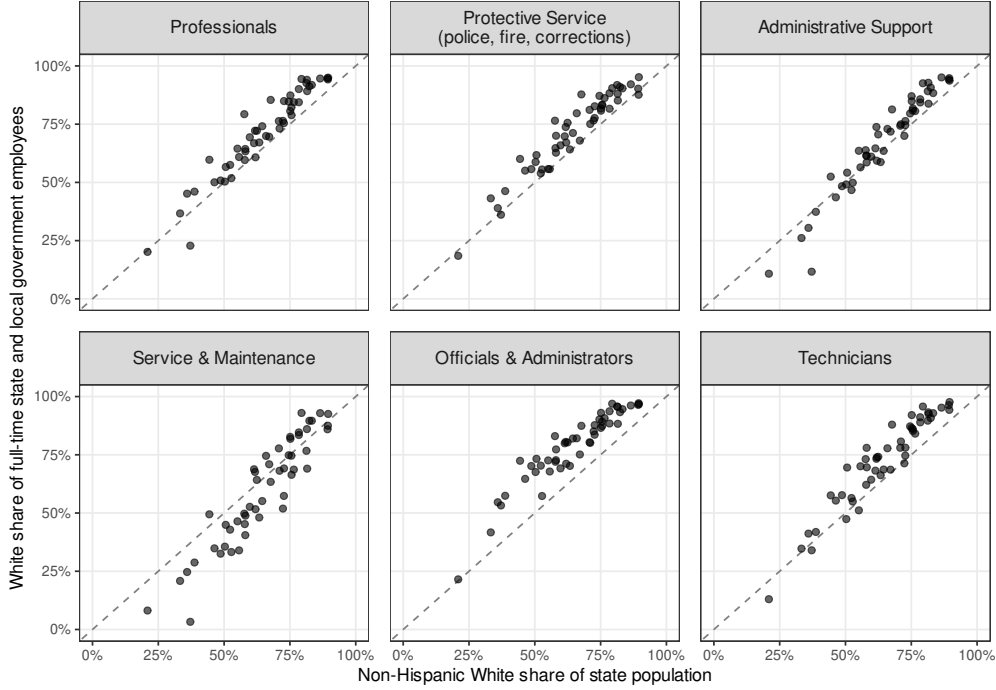
A growing body of evidence shows that many US state and local bureaucracies are unrepresentative of the jurisdictions they serve. Across 50 states and the District of Columbia, the workforce of state and local agencies is systematically Whiter than the resident population, with the gap most pronounced for officials and administrators (a median 14 percentage points), technicians (9 points), and protective service occupations such as policing, firefighting, and corrections (7 points; see Figure 1).¹ Policing has long been singled out for its skew. Across a sample of 98 major US law enforcement agencies, 56% of officers are White compared with only 38% of the population in the relevant jurisdictions (Ba et al., 2025). As policing issues have become increasingly polarized along partisan lines (Grosjean et al., 2023; Gaudette, 2025), scholars and commentators have also paid increasing attention to the fact that law enforcement officers are disproportionately drawn from Republican and conservative bases (Ba et al., 2025; Reny et al., 2025). 32% of officers identify with the Republican Party compared with 14% of voting-age citizens, while only 31% identify with the Democratic Party compared with 43% of civilians (Ba et al., 2025).

These representational gaps matter for several reasons. First, local governments are a major employer. In 2023, local jurisdictions across the US employed 14.1 million individuals and spent more than \$855 billion or 34% of their annual expenses on wages and salaries for employees (US Census Bureau, 2023, 2025).² This makes public employment a key channel of distributive spending and economic mobility, particularly for historically marginalized groups. Second, who staffs local government shapes both performance and legitimacy. Bureaucrats from underrepresented groups often deliver more equitable services to those groups (Bradbury and Kellough, 2011; Nicholson-Crotty et al., 2016; Ba et al., 2021; Harvey and

¹By aggregating state and local government employment to the state level, this comparison masks how the gap is distributed across local jurisdictions within a state. Appendix Figure A6 affirms that the same pattern holds at the level of Public Use Microdata Areas (PUMAs, approximately 100,000 residents). Across 2,462 PUMAs in the 2023 American Community Survey, state and local government workers living in a given PUMA are 3.9 percentage points Whiter at the median than that PUMA’s residents.

²This excludes expenses on retirement funds which also constitute a substantial source of expenditures.

Figure 1: Representativeness in US State and Local Government Employees



Note: Each point is one of the 50 states or the District of Columbia. The vertical axis is the White share of full-time state and local government employees in the indicated job category; the horizontal axis is the non-Hispanic White share of the state population. Points above the 45-degree line indicate workforces that are Whiter than the state population. Source: Census data from 2023 EEO-4 (EEOC) and 2023 American Community Survey 1-year estimates. EEO-4 covers state and local government employers with 100 or more employees. Public elementary and secondary school workforces are reported separately under EEO-5 and are not shown. State-level aggregation may mask within-state variation across local jurisdictions.

Mattia, 2022; Harris, 2023), partisan identification can predict how bureaucrats exercise discretion (Donahue, 2023; Lerman and Page, 2015), and the mere presence of demographic representation fosters citizens' trust and cooperation with the state (Ricucci et al., 2018; Miller and Segal, 2018). The dominance of certain racial and partisan groups is also striking against the institutional design of these bureaucracies. Most US local agencies operate under formalized civil service systems with merit-based examinations, standardized promotions, and tenure protections, which were established precisely to insulate bureaucratic composition from partisan influence and should, at least in theory, enable members from all groups and parties to win employment in the public sector (Skowronek, 1982; Kuipers and Sahn, 2023).

Existing explanations for representational gaps in formally meritocratic systems have largely treated civil service as a binary institution, asking whether merit-based hiring affects selection rather than how meritocratic rules play out within it. In this tradition, barriers at written entry exams have received the most attention, either through hidden curricula that reproduce inequalities via unequal access to test preparation and biased content (Portillo et al., 2020; Moreira and Pérez, 2025; Riccucci and Riccardelli, 2015) or through self-selection in who chooses to apply (Sidanius et al., 1994; Reny et al., 2025).

This paper instead treats meritocratic civil service as a system of successive stages, each governed by its own mix of formalized and discretionary procedures. This within-system view shifts attention in two respects. First, representation is determined at every stage of a career, beyond entry. Who is appointed from the pool of exam passers, who advances through the ranks, and who stays long enough to shape the agency all determine its composition. Second, the procedures governing these stages are not uniformly codified. The standardized written exams are paired with background screenings, discretionary promotions, and assignments, where far fewer criteria are fixed in advance. Because each stage filters the pool that enters the next, disparities arising along this sequence compound into the gaps observed in the bureaucracy.

Tracing where these disparities arise across a career has been hampered by data limitations, since it requires linking individual-level records across hiring, promotion, and attrition over many years. I use the New York City Police Department, one of the largest local bureaucracies and the largest local police force in the United States, as a case, and provide a first systematic decomposition of representational disparities across its career pipeline. I combine records from various administrative sources. Civil service lists from both active and terminated exam registers identify roughly 99,000 candidates who passed an entry-level exam between 2014 and 2021, including those never hired. City Record announcements, NYPD officer profiles, and payroll records then track the appointments, promotions, assignments, awards, and exits of 58,000 sworn officers. I link these to individual voter registrations, to

measure partisanship, race, and demographics at each stage.

The findings cut against the prevailing focus on written exams, showing that representation is determined less by standardized meritocratic procedures than by the discretionary parts of civil service careers.³ Although the application and written tests do produce some sorting by partisanship, they are insufficient to explain the representational gaps that ultimately emerge at the workforce level. Additional gaps emerge and compound at the discretionary procedures that follow, including post-exam character investigations and psychological evaluations, discretionary promotions and assignments, and involuntary terminations. Among applicants with comparable exam scores, Republican and White candidates are about 8 and 3 percentage points more likely to be appointed than Democratic and Black candidates. Additionally, Republican and White officers are more likely to be promoted to prestigious assignments, receive departmental awards, and remain with the agency longer and are more likely to leave voluntarily. A simple simulation shows that equalizing rates at the discretionary hiring procedures alone could increase the appointment of Democratic and Black candidates by 78% and 27%, respectively.

Throughout, I report differences in hiring, promotion, and exit rates between demographic groups both overall and among candidates or officers facing similar institutional conditions, i.e., taking the same exam, scoring similarly, or joining the same police cohort. These comparisons are descriptive by design, following calls for larger-scale quantitative description (Gerring, 2012). While description of this kind cannot identify or quantify mechanisms for these disparities, it can evaluate their observable implications, since competing accounts predict different patterns. One possibility is that Republicans and Whites advance because of policing-relevant qualities the written exam does not capture. This account is difficult to sustain. Unhired Black and Hispanic exam passers are nearly twice as likely to be appointed at the city’s other major law enforcement agency, hired White and Republican

³I focus on partisan and racial representation and set gender aside in the main analysis. Unlike race or partisanship, gender is subject to some formally sex-differentiated screens, such as physical tests, so post-exam gaps cannot be attributed to discretion as cleanly, and I leave a gender-specific treatment to future work.

officers have slightly worse disciplinary records, and observable quality gaps do not widen with tenure. Two channels fit the evidence better. Candidates with thinner ties to the agency and fewer resources appear to withdraw during a long and uncertain hiring process, which can take up to four years: Democratic and non-White candidates have fewer personal connections to serving officers and benefit less from connections they have, they more often sit other civil service exams, and non-hired passers cluster at lower incomes. At the same time, gaps in awards, coveted assignments and involuntary exits point to selection by the agency on its own notion of fit.

The paper makes several contributions. First, it reframes the study of representation in meritocratic civil service from a binary view of merit versus patronage to a within-system view that decomposes the pipeline by degree of formalization (Ruhil, 2003; Rauch and Evans, 2000; Dahlström and Lapuente, 2017; Kuipers and Sahn, 2023). The disparities it documents arise within the rank-and-file, absent the top-down political appointments that produce partisan bureaucracies elsewhere (Lewis, 2008). In this, the paper joins a growing literature on how bureaucracies themselves, rather than politicians, shape who is hired and promoted (Hassan et al., 2024; De Juan et al., 2026; Brierley et al., 2026). Second, it extends the literature on bureaucratic selection by analyzing entry together with the careers that follow, including promotions and reassignments. A growing body of work traces these later career procedures, examining promotions (Pierskalla et al., 2021; De Juan et al., 2026) and transfers (Brierley, 2020), but typically separately from entry, so their relative contribution to who ends up and stays in the bureaucracy remains unclear. Third, by applying the pipeline logic of the political selection literature (Fox and Lawless, 2011; Folke and Rickne, 2025) to bureaucratic careers and extending emerging evidence on partisan sorting in private-sector workplaces (McConnell et al., 2018; Chinoy and Koenen, 2024; Colonnelli et al., 2025) to a public-sector setting designed to prevent it, the paper bridges work on political pipelines and the politics of the workplace. The decomposition also offers reform-minded policymakers a clearer view of which selection processes to target. Diversification of police recruitment is

one of the oldest reform efforts in American policing, and decades of litigation and consent decrees on the written exams have made the first entry point the central target of reform (McCrary, 2007; Linos, 2017), thus largely neglecting later, more discretionary procedures in the career pipeline.

2 Civil Service, Discretion, and Disparities in Bureaucratic Careers

Civil service systems — now standard in most modern bureaucracies — rest on two key premises: political neutrality and meritocratic competence. In the US, the federal Pendleton Act of 1883 followed by numerous state and local jurisdictions throughout the nineteenth and early twentieth centuries established competitive exams and civil service commissions so that government jobs were awarded on the basis of merit rather than political connections (Kuipers and Sahn, 2023; Anzia and Trounstein, 2025). These reforms responded to a system widely seen as corrupted by partisan favoritism and machine politics, where political elites distributed coveted government jobs to political loyalists through patronage and direct appointments (Skowronek, 1982; Reid and Kurth, 1992; Ruhil, 2003). By removing staffing decisions from partisan control, civil service systems promise a bureaucracy insulated from the electoral interests of those in power. And by replacing discretionary appointments with standardized examinations, Weberian merit-based systems require that government employees are selected based on their ability and competence rather than other personal characteristics, such as race, gender or class (Weber, 1946).

Together, these premises, at least in theory, imply equal access to government jobs and upward social mobility. Weber himself recognized this function of competitive examinations, noting that they “mean or appear to mean a ‘selection’ of those who qualify from all social strata rather than a rule by notables” (Weber, 1946, p. 240). If government employment is determined by competence rather than connections, members of all groups, parties, and

classes should be able to compete for and advance through public service on equal terms (Kuipers and Sahn, 2023; Peters, 2015). An early generation of scholarship on US civil service reform largely confirmed that it delivered on this promise, eliminating de jure discrimination on grounds of partisan politics, race, and national origin (Van Riper et al., 1963; Hoogenboom, 1968). Similarly, recent work finds that reform did open avenues to representation for constituencies previously locked out of government jobs, especially in smaller municipal bureaucracies (Kuipers and Sahn, 2023). Nevertheless, in practice, local bureaucracies across US jurisdictions and agencies today remain dominated by certain partisan and racial groups. What explains these persistent representational disparities in professionalized bureaucracies governed by civil service rules and political insulation?

Existing scholarship offers two main accounts. The first emphasizes self-selection in recruitment. Individuals whose values align with an organization’s mission and culture are more likely to seek employment there (Schneider, 1987; Besley and Ghatak, 2005). In policing, for example, a growing body of work documents that those who aspire to law enforcement careers hold more conservative and authoritarian attitudes before they apply (Sidanius et al., 1994; Reny et al., 2025), and that recruitment messaging can shift who is drawn to apply (Linos, 2017). While this work highlights important supply-side dynamics, it focuses almost exclusively on who seeks to enter a profession, not on how institutional processes shape who is ultimately hired, promoted, or retained.

A second account focuses on the “hidden curriculum” of examination-based systems, showing how ostensibly neutral merit exams can reproduce existing societal inequalities. As Weber (1946, p. 240) put it, “democracy fears that a merit system and educational certificates will result in a privileged ‘caste.’” Written exams can disadvantage underrepresented groups due to unequal access to education, financial barriers to test preparation, or exam questions that reward general aptitude over job-relevant skills (Riccucci and Riccardelli, 2015; Portillo et al., 2020; Kuipers and Sahn, 2023; Moreira and Pérez, 2025). In fact, several scholars have argued that municipal civil service reforms in the US aimed to

advantage upper- and middle-class, native-born whites in the competition for government employment (Hofstadter, 1955; Shefter, 1993). Studying the implementation of civil service exams through the Pendleton Act, Moreira and Pérez (2025) find that the reform significantly reduced the representation of lower-status and immigrant groups while raising the share of employees with higher-status parental occupations in the US federal bureaucracy. These concerns are especially pronounced in uniformed services, where written exams remain a dominant selection tool and have been subject to repeated litigation and consent decrees since the 1960s. As Riccucci and Riccardelli (2015, p. 352) conclude, “it is the use of written exams that has perhaps been the biggest obstacle for police and fire departments to achieve social diversity based on race and ethnicity.”

2.1 Decomposing Civil Service Across the Bureaucratic Pipeline

While these accounts offer valuable critiques of exam-based selection, both treat the composition of a bureaucracy as primarily determined at the entry point. Yet, bureaucratic careers extend well beyond the first selection stage. Who is appointed from the pool of exam passers, who advances through the ranks, and who stays long enough to shape the organization also determines the composition of the bureaucracy. The existing literature on the effects of civil service systems has largely treated meritocracy as a binary system-level variable: a jurisdiction either has competitive civil service exams or it does not (Rauch and Evans, 2000; Folke et al., 2011; Portillo et al., 2020; Kuipers and Sahn, 2023; Moreira and Pérez, 2025). In contrast, I argue that civil service is better understood as a pipeline of career stages with varying degrees of formalization. Decomposing this pipeline — rather than focusing on the mere existence of merit exams — is necessary to understand where representational disparities emerge and how much each stage contributes.

The key logic of civil service systems is that formalized rules tie career outcomes to codified, measurable criteria. Yet, as Figure 2 illustrates, within most professionalized civil service systems, the procedures that determine career outcomes vary considerably in how far

they rely on such criteria.⁴ Some procedures are highly formalized. Written entrance exams, for instance, typically define clear cutoff scores necessary to be placed on official civil service lists, and agencies then hire among the top-scoring candidates — a provision commonly known as the “rule of three.” Similarly, competitive promotions require candidates to pass exams and be placed on rank-ordered lists. Across these formalized procedures, criteria are specified in advance, outcomes are determined by measurable performance, and the process is often administered by an independent body (such as a civil service commission) rather than the hiring agency itself.

Figure 2: Decomposing the Civil Service Career by Stage and Degree of Formalization

	Application	Entry	Promotion	Exit
Formalized <i>codified criteria</i>	eligibility criteria	competitive written exam, ranked list	promotion exams, ranked list	retirements & resignations, tenure & pension rules
Discretionary <i>evaluator judgment, self-selection</i>	self-selection (who applies)	background & psychological screening, long wait times	discretionary promotions, awards, elite unit assignments	dismissals & terminations



representational disparities compound across stages

Note: Illustration of the argument. Columns are sequential career stages; the two rows distinguish formalized procedures, which tie outcomes to codified criteria, from discretionary procedures, which leave more room for both evaluator judgment and candidate self-selection. Every stage combines the two, and the argument expects disparities to concentrate in the discretionary band (shaded) and compound across stages.

Other procedures are far less constrained. Performance criteria in civil service systems tend to grow more subjective at higher ranks, as measurement becomes more difficult and more dependent on cultural and social values (Peters, 2015). Assignments to specialized units, nominations for awards, and promotions between grades that do not require competitive exams lack standardized benchmarks. Decisions about discipline and involuntary

⁴A parallel literature on street-level bureaucracy highlights the importance of discretion in how public servants deliver services to citizens (Lipsky, 1980). This paper examines discretion in personnel processes that determine who those street-level bureaucrats are in the first place.

separation are similarly ambiguous. And even within the entry stage, formalized and discretionary procedures often coexist: a standardized written exam may be followed by a character investigation or psychological evaluation with considerable subjective judgment.⁵ To be sure, as the hidden curriculum literature has demonstrated, formalized procedures are not immune to producing disparate outcomes. But we know little about where across the pipeline these disparities are produced, and how much the less formalized procedures contribute relative to the exam. If discretionary procedures increase the number of factors that shape career outcomes and these disadvantage underrepresented groups, we should expect these disparities to concentrate there across the pipeline.

This expectation is consistent with both demand-side and supply-side mechanisms. On the demand side, discretionary procedures give institutional gatekeepers greater freedom to exercise preferences (whether conscious or not) in ways that formalized criteria would constrain. On the supply side, less formalized procedures tend to be longer, less transparent, and harder for candidates to navigate, creating more opportunities for differential self-selection. Candidates with weaker networks, fewer outside resources, or less familiarity with organizational norms may be more likely to withdraw or underperform in unstructured settings. While it is hard to empirically adjudicate between these mechanisms, decomposing the civil service pipeline by stage and by degree of formalization shows where the system’s neutrality premise bites and where it does not, which matters for institutional design regardless of the underlying mechanism.

2.2 Formalization and Discretion in Policing Careers

US law enforcement is a good case to test this argument — that the representational effects of meritocratic civil service differ across the formalized and discretionary parts of the career

⁵A large literature on private-sector organizations documents that discretionary procedures (e.g., subjective evaluations, unstructured interviews, informal networks) widen the set of factors that shape who is selected and promoted, while formalized procedures narrow outcomes to codified criteria (Reskin, 2003; Castilla and Benard, 2010; Rivera, 2012). These dynamics have received little attention in public bureaucracies.

— for several reasons. First, in most local jurisdictions, policing is governed by strong civil service systems (Ricucci and Riccardelli, 2015; Ornaghi, 2019; Anzia and Trounstone, 2025). Most US states decentralize civil service procedures to the local level, and nearly all major cities administer written exams for police hiring and promotion, even where statutes do not formally require them and even in states that have weakened or abolished state-level civil service (e.g., Florida, Georgia, Texas) (Ricucci and Riccardelli, 2015).

Second, US law enforcement remains systematically unrepresentative of the jurisdictions they serve, and a robust empirical literature shows that who officers are shapes how they do their jobs. Ba et al. (2021) find that Black, Hispanic, and female officers in Chicago use force less often than White and male officers on the same shift and patrol assignment, especially against Black civilians and in majority-Black areas.⁶ There is also some evidence that Democratic officers make fewer traffic stops and arrests, use force less often, and exhibit smaller racial disparities than Republican officers (Donahue, 2023).⁷

Third, policing careers run through a structured pipeline with substantial variation in the degree of formalization. For example, standardized written entry exams with pre-specified cutoffs are paired with more discretionary screens, including character investigations, psychological evaluation, and physical tests. Similarly, while promotions to civil service ranks such as sergeant or lieutenant rely on formalized exams, horizontal moves, such as grade changes within the detective track or assignments to elite units, are far more discretionary. This within-agency variation allows me to compare formalized and discretionary career stages while holding organizational culture and institutional context largely fixed.

⁶This does not imply that increasing passive representation will mechanically reduce racial disparities in policing outcomes. Work on perceptions of representativeness and the role of local context (Brunson and Gau, 2015; Benton, 2020; Socia et al., 2021) suggests the link between representation and performance is complex, and more work is needed on how macro-level socioeconomic and institutional factors interact with officer identity.

⁷Ba et al. (2025) find little difference between Republican and Democratic officers in Chicago and Houston after multiple-comparison corrections.

3 Case and Data

I study the New York City Police Department (NYPD), the largest municipal police force in the United States. NYC operates one of the most formalized civil service systems in the country, with strict rules governing hiring, promotions, and separations. Critically for this paper, the NYPD career pipeline contains both highly formalized and highly discretionary procedures, and I am able to observe each in the administrative data. At entry, I observe both exam scores and ultimate hiring outcomes, allowing me to compare who passes the written exam with who is ultimately appointed after the discretionary screening process. For career progression, I can distinguish formalized promotions (exam-based advancement to sergeant, lieutenant, and captain) from discretionary ones (grade changes within the detective track, elite unit assignments, and departmental awards). And for attrition, I observe the reason for each officer’s separation, distinguishing voluntary retirements from involuntary exits such as dismissals and terminations. Finally, NYC’s elected officials have been overwhelmingly Democratic for decades. Showing the dominance of Republican and White employees in a largely liberal city government makes the case particularly useful for studying how these disparities emerge within the bureaucratic pipeline itself rather than through political turnover and influence.

I start with a roster of roughly 300,000 unique employees across the six major agencies of NYC in terms of their staff size, including the Department of Education, the Police Department, the Fire Department, the Department of Correction, the Department of Social Services, and the Department of Sanitation. These data comes from the NYC annual payroll between 2014 and 2021 and covers employees with appointment dates between 1970 and 2021.⁸ To identify the various demographic attributes of these employees, I merge the employment records with the 2021 L2 voter file based on employees’ last names, first names,

⁸<https://data.cityofnewyork.us/City-Government/Citywide-Payroll-Data-Fiscal-Year-/k397-673e>; I identify unique employees based on the last name, first name, middle initial, and appointment date.

and middle initials. I restrict possible matches to registered voters in the city’s five boroughs or one of the neighboring counties of New York State since NYC agencies require their employees to reside within these areas.⁹ Following related work (Ba et al., 2025), I employ the probabilistic record linkage algorithm by Enamorado et al. (2019) and retain all matches with a posterior match probability of at least 0.7.

To study the specific drivers of selection among NYPD officers, I then add information on the career trajectories of about 58,000 uniformed police employees, including appointments, promotions, and retirements, from official records published in the City Record newspaper since 2014.¹⁰ For a cross-section of 33,000 active officers (as of October 2021), I am further able to add information on their exact assignment, their arrest history, as well as their departmental awards from the official NYPD profiles.¹¹ Finally, I obtain data on roughly 99,000 entry-level and 5,700 promotion exams for the NYPD between 2014 and 2021 from active and terminated civil service lists¹², which I match to the L2 voter file and the NYPD officers on the payroll. This allows me to assess the attributes of both hired and non-hired NYPD aspirants.¹³ I link these administrative data probabilistically based on individuals’ full names and other employment details, where possible, in all these merging procedures. Appendix A describes these different matching procedures in more detail. As Figure A1 shows, I can correctly match most employees and records with a very high probability—the median posterior probability of a match is above 0.95 across all matching procedures.

To measure the partisanship and race of individuals in my data, I rely on the information in the L2 voter files. For partisanship, I focus on the three main categories included in L2:

⁹Technically, NYC also allows employees to reside in Putnam and Orange County. Yet, because these counties are further away from NYC boundaries and relatively small, I exclude these to reduce the risk of false positive matches.

¹⁰<https://www.nyc.gov/site/dcas/about/cityrecord-editions.page>

¹¹<https://nypdonline.org/link/2>

¹²https://data.cityofnewyork.us/City-Government/Civil-Service-List-Active-/vx8i-nprf/about_data, https://data.cityofnewyork.us/City-Government/Civil-Service-List-Terminated-/qu8g-sxqf/about_data, <https://www.nyc.gov/site/records/nyc-government-records/publication-s-portal.page>

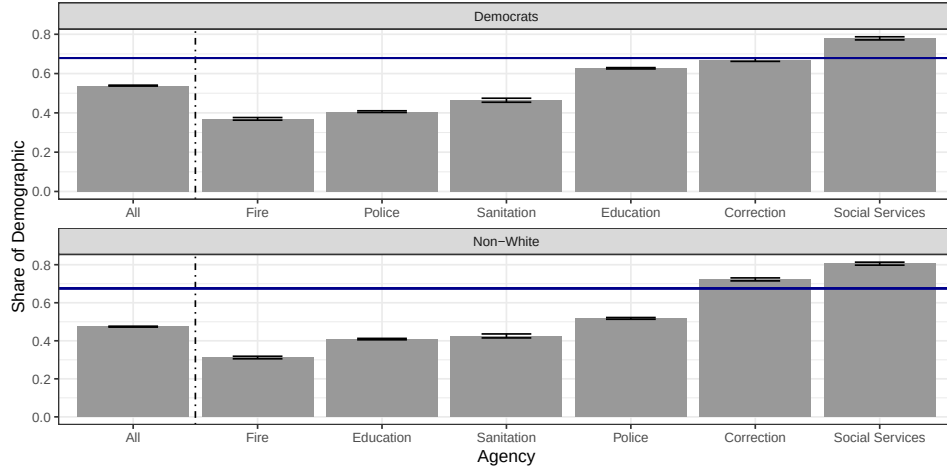
¹³Note that the exam data does not provide information on *all* applicants, but instead contains individuals who successfully passed the exam (i.e., those with a score of 70/100 or above).

Democrat, Republican, and Non-Partisan. Together, these comprise 96% of the 7,940,000 voter registrations in NYC and its surrounding counties. Note that the L2 information on partisanship in New York — unlike for other US states — is based on official registration records and does not require imputations. However, I must rely on L2’s proprietary imputation algorithm to measure bureaucrats’ and citizens’ race. To code racial categories of registered voters, L2 combines surname information with census-block demographics in the spirit of standard Bayesian Improved Surname Geocoding (BISG) (Imai and Khanna, 2016). To assess its validity, I compare my estimates against (1) official counts of race groups reported by the NYPD to the 2020 Law Enforcement Management and Administrative Statistics (LEMAS) survey and (2) measures of employees’ race using Bayesian predictions (Imai and Khanna, 2016) in Appendix A.7. Overall, the results suggest that, if anything, I likely *underestimate* the share of White groups in the NYPD, and the true representational gaps between Whites and non-Whites may be even larger than reported. Among non-White employees, the L2-based estimates have a disproportionately higher share of “Other/unknown race.” If these unknowns are predominantly Black, Hispanic, or Asian, the gaps for specific groups (especially Black) may be smaller than reported here, though this does not affect the overall White-vs-non-White comparison.

4 Political Sorting in NYC’s Bureaucracy

I begin by comparing the demographic composition of front-line employees (uniformed and public-facing staff plus agency leadership, excluding administrative personnel) across NYC’s six major agencies with the city’s voting population. Figure 3 displays results along partisanship and race. Most agencies diverge from their jurisdictions, albeit in different directions. The police, fire, and sanitation departments have substantially smaller shares of Democratic and non-White members than the NYC electorate. The NYPD underrepresents Democrats by 28 percentage points and non-Whites by 15 percentage points. The Department of Correction, in contrast, closely matches the city’s voters in partisanship and even *overrepresents*

Figure 3: Share of Demographics, Agencies vs. Registered Voters in NYC



Note: The bold line represents respective share of demographic among NYC voters. Agency employees only include uniformed/public facing employees plus leadership (i.e., I exclude administrative and other agency staff). Agency personnel includes individuals living in NYC’s five boroughs and neighboring boroughs, while NYC citizens are restricted to NYC’s five boroughs. Agency estimates are weighted by the posterior probability of matches between agency payrolls and the voter file.

non-Whites — a striking difference given that both are law enforcement agencies operating under the same civil service system. The Department of Education and the Department of Social Services skew more Democratic and non-White than the general population. Representational gaps in NYC’s police force also vary geographically: Democrats are underrepresented and Republicans overrepresented across all NYC boroughs, with racial gaps particularly stark in high-crime and majority-Black communities (see Figures A7 and A8).

5 Tracing Stages in the NYPD’s Pipeline

What accounts for the dominance of White and Republican employees at NYC’s police department? While I focus on partisan and racial representation gaps across the pipeline, I acknowledge that these are bundled categories. Fixed effects and controls allow me to make more informative comparisons and to account for measurable differences across groups, for instance, by comparing officers within the same entry cohort or with similar exam scores. However, some differences remain unobservable, such as public sector motivation, previous

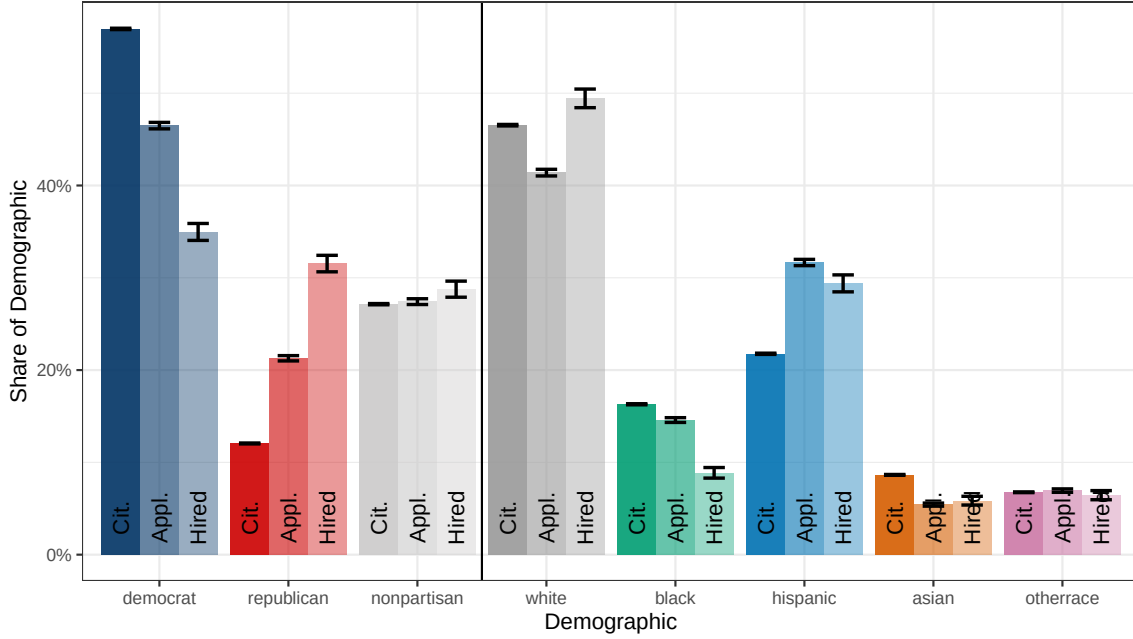
experiences, or biases by selectors. Importantly, these unobserved factors may themselves be part of the process producing these gaps rather than an alternative explanation for it. The goal of the decomposition is not to identify causal mechanisms for representational gaps but to show where gaps concentrate and how much each stage contributes.

5.1 Selection Into the Force

The NYPD hiring process combines both formalized and discretionary procedures. To become a police officer at the NYPD, applicants need to be between 21 and 35 years old, have earned 60 college credits with a minimum GPA of 2.0 or 2 years of military service and must live within NYC or one of the neighboring boroughs at the time of appointment. Candidates who satisfy the basic selection criteria first take a written exam offered by the Department of Citywide Administrative Services (DCAS) which covers 9 cognitive abilities. Only candidates with a minimum score of 70% are placed upon an eligible civil service list for appointment, with better performing candidates being placed further up on the list. After an eligible list is certified by the DCAS commissioner, agencies can appoint from the list in ranking order. This regulated process is followed by several additional screens administered directly by the NYPD, which vary in how much discretion they leave to evaluators. Once a list number is reached, candidates undergo objective medical, physical, and drug-and-alcohol tests with codified pass/fail standards, alongside more discretionary written and oral psychological evaluations and a “character and satisfactory background” investigation that screens for arrest records, convictions, and discharge of employment. After successful completion of this process, candidates are then hired as probationary officers and are required to complete the 6-month police academy.

To follow the process of hiring at the NYPD empirically, Figure 4 shows the share of party and racial categories among three different groups: (1) voters in counties eligible for

Figure 4: Eligible Citizens, Police Exam Passers, Hired Exam Passers - NYPD



Note: The three bars among each partisan and racial group represent (from left to right) (1) share among NYC voters, (2) share among police exam passers, and (3) share among hired exam passers. Voters and exam passers are matched on age. All estimates are weighted by posterior probability of a match (L2 and exam data).

employment at the NYPD¹⁴, (2) all candidates who passed the NYPD entrance exam¹⁵, and (3) those applicants who were successfully hired and appointed to the police academy. Self-selection into policing and screening through the written entry exam can explain some of the partisan, though not racial, sorting at entry. At the exam stage, the applicant pool already underrepresents Democrats relative to eligible voters (47% vs. 57%) and overrepresents Republicans (21% vs. 12%). The racial pattern, however, runs in the opposite direction. The share of Black candidates almost matches the eligible pool (16% and 15%), Hispanics are overrepresented among exam passers (32% vs. 22%), and White candidates are *underrepresented* (41% vs. 47%). Because Democrats vastly outnumber Republicans among eligible voters, the partisan share gap reflects a Republican-to-Democrat application rate ratio of

¹⁴I rely on age and residency information in the L2 voter data. For education, I use Bayesian imputation with census data from the 2019 American Community Survey based on the census tract.

¹⁵Unfortunately, the exam data only provides individuals placed on eligible civil service lists and does not provide the pool of unsuccessful candidates.

roughly 2.3 to 1, while the Black-White-gaps at this stage are close to parity in rate-ratio terms.

Table 1: Differences in Hiring By Exam Passer Characteristics

	Model 1	Model 2	Model 3	Model 4
Republican	0.09*** (0.00)		0.08*** (0.00)	0.07*** (0.00)
Non-Partisan	0.04*** (0.00)		0.03*** (0.00)	0.03*** (0.00)
Black		-0.06*** (0.00)	-0.03*** (0.00)	-0.01*** (0.00)
Hispanic		-0.02*** (0.00)	0.00 (0.00)	0.01** (0.00)
Asian		-0.00 (0.01)	0.02** (0.01)	0.02** (0.01)
Other Race		-0.01** (0.00)	0.00 (0.01)	0.01 (0.01)
Exam Score Q2 (within exam)	0.01** (0.00)	0.01* (0.00)	0.01* (0.00)	0.01* (0.00)
Exam Score Q3 (within exam)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)
Exam Score Q4 (within exam)	0.03*** (0.00)	0.03*** (0.00)	0.02*** (0.00)	0.03*** (0.00)
Exam Score Q5 / top (within exam)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)
NYPD Exam Cohort FE	Yes	Yes	Yes	Yes
Additional controls	No	No	No	Yes
Mean of DV	0.13	0.13	0.13	0.13
Adj. R ²	0.04	0.03	0.04	0.04
Num. obs.	75134	72851	69406	64340

Linear probability regressions, weighted by the posterior probability of a match between exam data and voter file, and between exam data and appointments. Exam performance enters through exam-specific score quintiles. NYPD Exam Cohort FE are fixed effects for candidates who sat the same exam. Additional controls include gender, age, and county of residence. HC1 standard errors are in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

The discretionary hiring process adds important further disparities on both dimensions. Democrats make up only 35% of hired candidates, and Republicans account for 32% of suc-

successful applicants. Similar trends are observed for race, where the share of Whites increases by 9 percentage points between the application and the hiring stages. These higher hiring rates for Republican and White applicants are not explained by exam performance.¹⁶ As Table 1 shows, among candidates of the same exam and similar scores, Republicans and White candidates are about eight percentage points and three percentage points more likely to be appointed than Democratic and Black applicants, respectively (Column 3);¹⁷ both gaps persist after observable controls, though the Black-White gap shrinks further to about one percentage point (Column 4). Hispanic and Asian candidates show no within-score hiring disadvantage relative to Whites. For Hispanic candidates in particular, the substantial application advantage seen in Figure 4 does not translate proportionally into hires, but this attenuation seems accounted for by lower exam scores rather than by discretionary screening. Figures A9 and A10 further indicate that these gaps persist across the entire score distribution, and that among hired exam-takers, Blacks and Democrats do not have substantially higher scores. Unlike work on women navigating barriers to political office, where those who break through are often positively selected on observable qualifications (Anzia and Berry, 2011; Ashworth et al., 2024), the underrepresented candidates who clear the NYPD’s discretionary screens do not seem stronger on measurable ability than their Republican and White counterparts.

If exam performance does not explain who clears the post-exam hiring stages, what does? In Section 6 I examine several candidate mechanisms and show that the gap is not well explained by differences in policing-relevant ability. It appears most consistent instead with differential self-selection during the long, discretionary post-exam process, shaped by candidates’ outside options, financial resources, and personal networks in law enforcement.

¹⁶Note that White candidates achieve slightly higher scores on the exams (see Table A6).

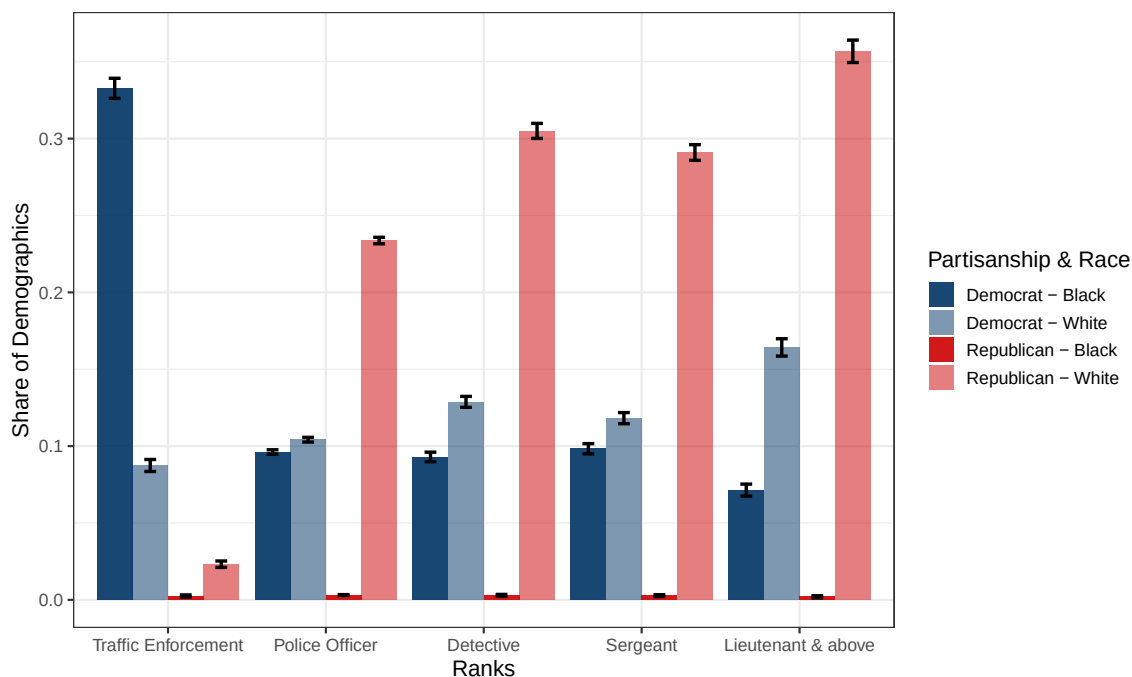
¹⁷As Table A7 shows, these results are highly robust to different functional form specifications for exam score controls.

5.2 Career Progression

Career progression at the NYPD, like its hiring process, involves both formalized and discretionary procedures: exam-based promotions to civil service ranks on the one hand, grade changes within the detective track, elite-unit assignments, and departmental awards on the other. The within-pipeline contrast that emerged at entry reappears in career progression.

Figure 5 provides a cross-sectional starting point. Using data on all uniformed employees and traffic enforcement agents on the NYPD payroll (2014–2021), it indicates that Democrats, particularly Black Democrats, are clustered among the lower ranks. For instance, among traffic enforcement agents 33% are Black Democrats, whereas only 6% of Lieutenants or above are Black and Democratic.¹⁸ For White Republicans, in contrast, these figures amount to 2% and 35%, respectively.

Figure 5: Share of Demographics by Police Rank



¹⁸Traffic enforcement agents (TEAs) are a separate civil service title with their own entry exam and hiring track, not a lower step on the police-officer career ladder. The TEA-to-lieutenant comparison thus reflects both between-track sorting at hiring and within-track promotion. The regression analyses in Table 3 restrict to sworn officers.

Table 2: Differences in Promotions by Officer Characteristics

	Official Promotions			Title Changes		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Republican	−0.00 (0.00)	−0.00 (0.00)	−0.00 (0.00)	0.01* (0.00)	0.01 (0.00)	0.00 (0.00)
Non-Partisan	−0.00 (0.00)	−0.00 (0.00)	−0.00 (0.00)	−0.00 (0.00)	−0.00 (0.00)	−0.00 (0.00)
Black	−0.01 (0.00)	−0.01 (0.00)	−0.01 (0.00)	−0.03*** (0.01)	−0.02*** (0.01)	−0.03*** (0.01)
Hispanic	−0.01** (0.00)	−0.01** (0.00)	−0.01** (0.00)	−0.02*** (0.00)	−0.02*** (0.00)	−0.02*** (0.00)
Asian	0.04*** (0.01)	0.03*** (0.01)	0.04*** (0.01)	−0.00 (0.01)	−0.01 (0.01)	−0.00 (0.01)
Other Race	0.03*** (0.01)	0.03*** (0.01)	0.03*** (0.01)	−0.00 (0.01)	−0.01 (0.01)	−0.01 (0.01)
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Exposure FE	Yes	Yes	Yes	Yes	Yes	Yes
Age, Gender	No	Yes	Yes	No	Yes	Yes
CCRB complaints	No	No	Yes	No	No	Yes
Mean of DV	0.09	0.09	0.09	0.17	0.17	0.17
Adj. R ²	0.10	0.10	0.11	0.19	0.19	0.20
Num. obs.	45990	43573	43573	45990	43573	43573

Linear probability regressions, weighted by the posterior probability of a payroll and voter file match. Level of observation: Uniformed employee. Outcome: Dummy for whether the employee received a promotion/title change between 2014 and 2021. All models control for the duration an employee is observed on the payroll since 2014 (Exposure time). Some models also control for age at appointment and gender, as well as total number of CCRB complaints over an officer’s full tenure. HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Yet, since these snapshots do not account for compositional differences across cohorts, Table 2 estimates disparities in promotion probabilities across partisanship and race within the same cohorts. Models (1)–(3) include only official promotions that require a promotion exam (i.e., sergeant and above), Models (4)–(6) further include discretionary promotions such as detective grade changes. The estimates do not provide clear evidence that Republicans and Whites are systematically more likely than Democrats or Black officers to receive *official*

promotions (Table 2, Columns 1–3).¹⁹ Two further checks reinforce this null result for the formalized, exam-based promotions. First, Table A11 shows that scores on the promotion exams are a strong predictor of receiving a promotion, and once exam performance is accounted for the small promotion gaps subside, consistent with Republicans and Whites performing slightly better on promotion exams rather than benefiting from differential treatment after the exam. Second, Table A10 indicates no substantial gaps in promotional aspirations, with Republican and White officers no more likely to take a promotion exam throughout their tenure than Democrats and non-Whites.²⁰

However, the racial gaps in career progressions are larger and significant when incorporating discretionary promotions and grade changes. Whites are estimated to be 3 and 2 pp more likely to receive both horizontal and vertical career movements than Blacks and Hispanics, respectively. With an overall promotion rate of 17%, these estimates are non-trivial. The partisan gap in promotions is small and sensitive to age and gender controls, but emerges more clearly in outcomes with even less structure. Republican and White officers are more likely to receive departmental awards and to be assigned to prestigious elite units, including anti-terrorism and special forces (Tables A12 and A13). Similar discretion in team assignments produces parallel disparities: teams headed by non-White (Democratic) leaders have lower shares of lower-ranked White (Republican) members and higher racial diversity overall (Tables A14 and A15).²¹

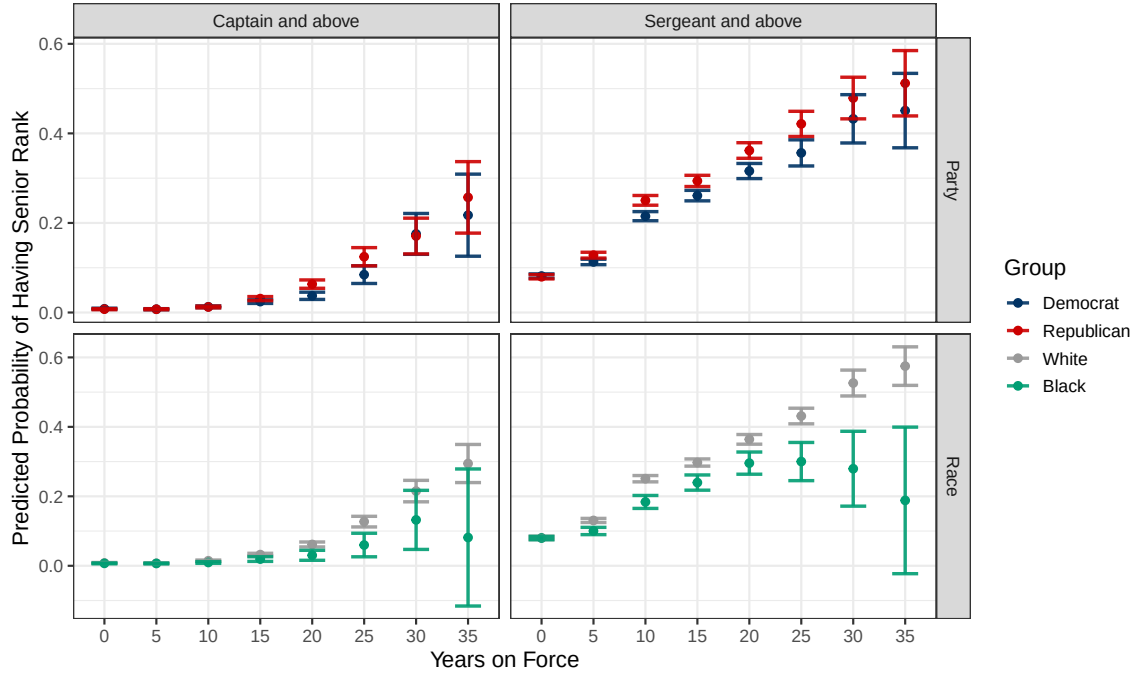
Figure 6 further assesses whether this partisan and racial seniority gap persists across officers’ tenure. The Black-White gap endures and widens over time, whereas Democrats seem to catch up to the ranks of Republicans after 30 years on the force. Taken together, this suggests that Republican and especially White officers benefit from steeper progress through the discretionary procedures of career advancement in law enforcement.

¹⁹Coefficients for other racial groups are small but cut in different directions, with Hispanic officers about 1 percentage point less likely and Asian and other-race officers 3–4 percentage points more likely than Whites to receive an official promotion.

²⁰The exception is Asian officers, who are more likely to take promotion exams.

²¹These analyses use a cross-section of active officers (as of October 2021) for whom more detailed information on assignments and awards is available.

Figure 6: Seniority Gap by Years of Experience

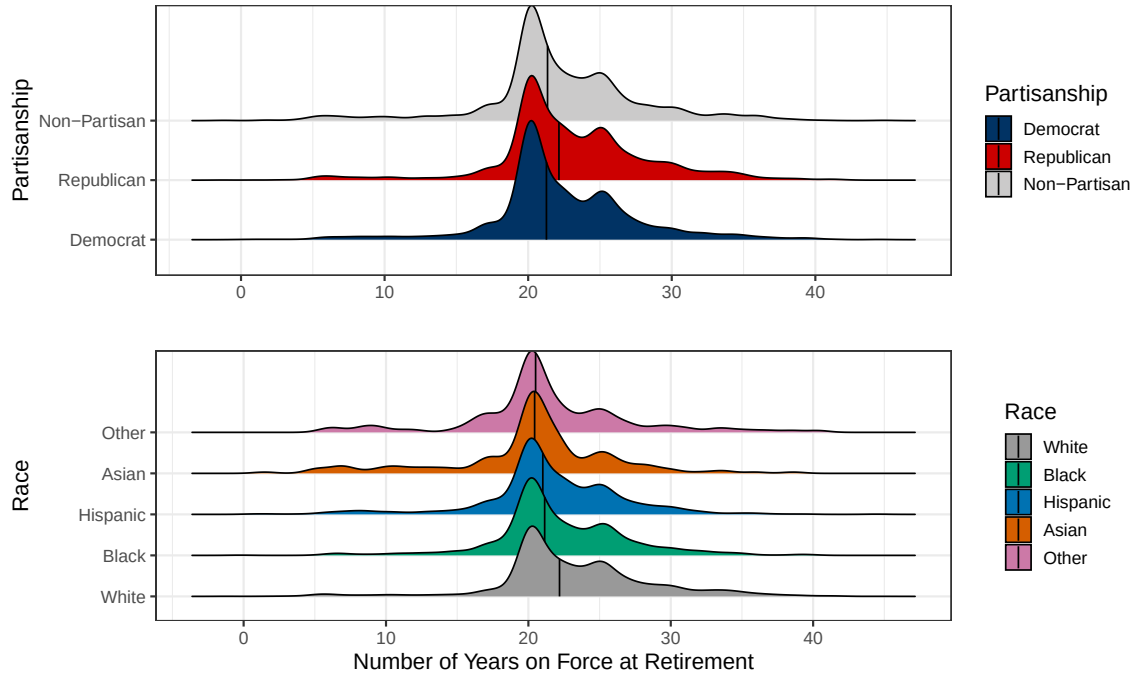


Note: Depicted are predicted probabilities of having a senior rank, with 95% HC1 confidence intervals. All underlying regression models (LPM) include officer cohort fixed effects.

5.3 Attrition

How do these differences in career trajectories translate into attrition? Like earlier stages of the pipeline, attrition has both formalized and discretionary procedures. Voluntary retirements at the 20-year mark are tied to formal pension eligibility, whereas dismissals and terminations are discretionary agency actions. Figure 7 depicts the distribution of years on the force at the time officers retire. Evidently, most officers retire around 20 and 25 years of service, when they become eligible for different retirement packages at NYPD. Yet, Republicans and White employees stay on the force slightly longer, often working beyond the retirement age of 20 years on the force. For instance, the median retiring Republican or White officer worked for 22.1 years, compared to 21.3 years for Democratic officers and 21 years for non-White officers. As Table A16 shows, these results hold when accounting for officers' age at appointment.

Figure 7: Distribution of Years on Force at Retirement

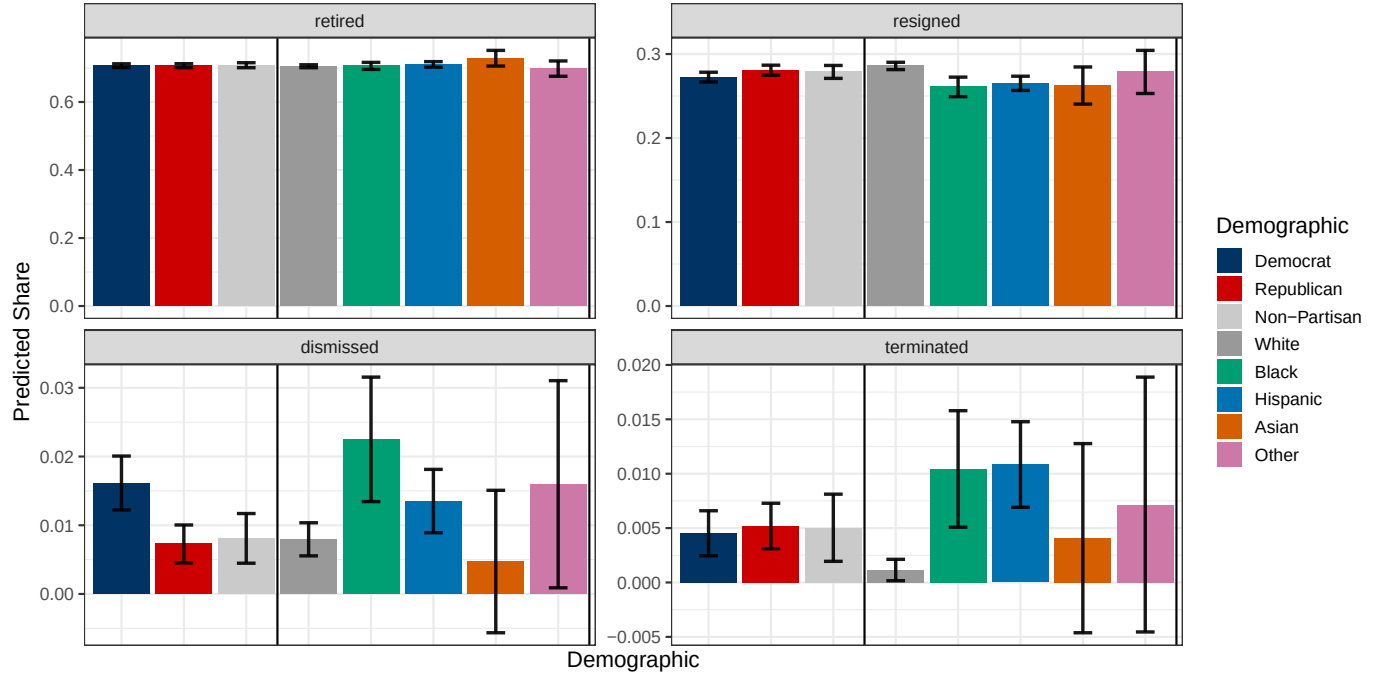


Note: Black lines indicate median years on the force at retirement.

The discretionary agency side of attrition shows the sharpest demographic gaps. Across all types of officers, retirements and resignations account for the bulk of exits (Figure 8). But involuntary exits (dismissals and terminations), where supervisors and the department decide who is removed, account for significantly higher shares of exits among Democratic, Black, and Hispanic officers than among Republicans and Whites. Specifically, 2% of exiting Democrats, 3.3% of Blacks, and 2.4% of Hispanics leave involuntarily, compared to 1.3% of Republicans and 0.9% of White officers. The pattern is most pronounced among early-career officers (Figures A17 and A18), where tenure protections are weakest.²²

²²Voluntary resignations show no similar disproportionate attrition rate for Democrats or Non-Whites, suggesting the early-career pattern reflects agency-side removal decisions rather than officer-side decisions to leave.

Figure 8: Predicted Probability of Exit Type



Note: The estimates are obtained from regressions of exit type (conditional on exit) on group demographic and cohort FE. The covariates are fixed at their observed values for predictions.

6 Mechanisms

The decomposition of the bureaucratic career pipeline of the NYPD shows that partisan and racial gaps emerge well beyond the formal written entry exam, at the discretionary procedures of hiring, promotion, and exit. What mechanisms could plausibly produce these gaps? Mechanisms are causal quantities and the descriptive design of this paper cannot adjudicate them fully. Yet, the evidence is informative, as candidate mechanisms imply distinct observable patterns, and the findings favor some accounts over others.

6.1 Sorting and Self-Selection

A first candidate account is sorting on the supply side. One version is *pre-entry self-selection*, where minority and Democratic candidates are less likely to apply, for instance because of distaste for policing or anticipated costs of the process. The application gaps documented

in Figure 4 are consistent with this. Yet, a pure pre-entry self-selection story has trouble explaining what comes after. Candidates among underrepresented groups who enter despite higher costs should be positively selected on persistence, motivation, and underlying ability, which would predict higher rather than lower hiring rates among non-Whites and Democrats (Ashworth et al., 2024). The hiring disadvantage for these groups at the end of the entry process, together with the fact that hired non-Whites and Democrats are not disproportionately high-scorers Figure A10, cuts against this reading.

Post-exam self-selection is more compatible with what we see. During the long, discretionary post-exam process, minority and Democratic candidates, especially those with better outside options, may exit at differential rates, and four features of the process make this likely.²³

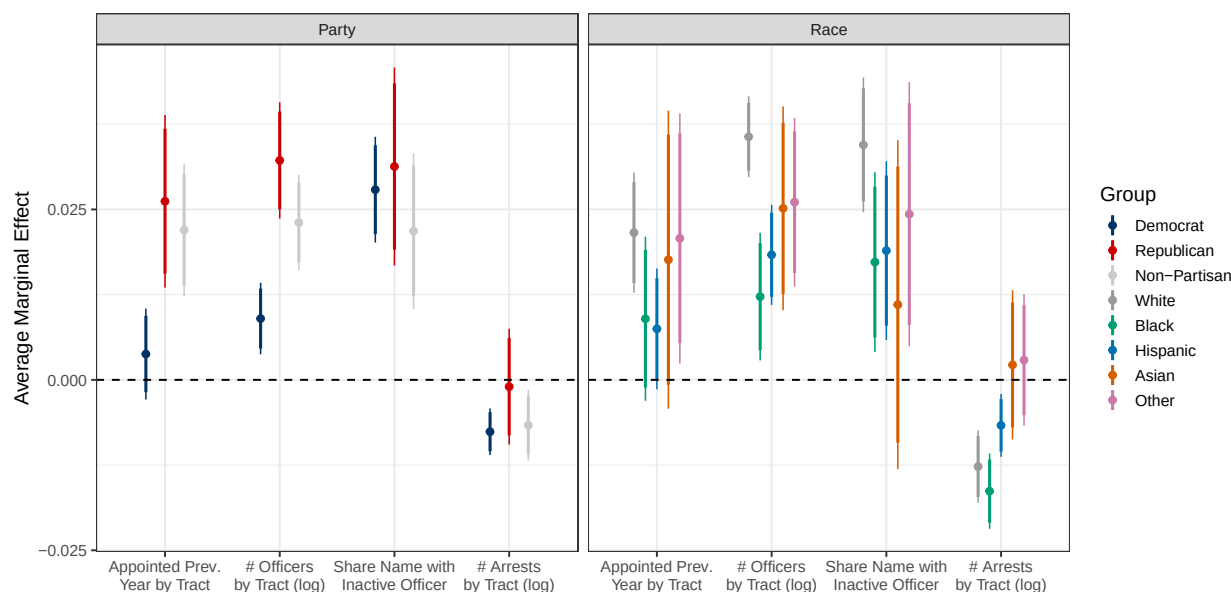
First, the process is unusually long and disqualification-heavy. Candidates can wait up to four years to be sworn into the academy, far longer than at peer agencies such as the Department of Corrections (Figure A11), and most of this reflects how long the NYPD waits before calling candidates from the eligible list (a median of 948 versus 282 days). NYPD candidates are also more likely to be disqualified during the psychological, medical, and character screens (Figure A12).

Second, personal networks in law enforcement shape who is hired, and their returns are unequally distributed. Using applicants' addresses and surnames, I measure each candidate's connection to NYPD personnel through the density of current officers in their census tract and surname matches with retired officers. These networks are denser among Whites and Republicans. For example, Whites (Republicans) live in census blocks with an average of 31 (33) active police officers, compared to only 24 (23) for Democrats (non-Whites). Additionally, the advantages of personal connections are markedly stronger for White and Republican applicants (Figure 9). For these groups, a doubling of the number of current NYPD officers residing in their neighborhood correlates with a 3-percentage-point increase in hiring proba-

²³Requests for data on the specific reasons and timing of attrition from the hiring process have either been denied or remained unanswered (FOIL-2024-134-00069, FOIL-2024-056-23791, FOIL-2023-056-02128).

bility — a 30% relative gain off a 10% base rate. Democratic and Black applicants show much smaller and often negligible benefits from such neighborhood connections, and instead face marginally lower hiring probabilities when living in tracts with higher arrest rates. These findings suggest that informal networks may contribute to the observed disparities in who successfully navigates the NYPD hiring process.

Figure 9: Personal Networks and Hiring Probabilities



Note: All regressions include exam and score bin fixed effects. Models (1), (2), and (4) further control for the population size per tract. Model (3) controls for name commonality using 2010 census data.

Third, underrepresented candidates appear to seek and use more outside options. Democratic and non-White candidates are more likely than Republicans and Whites to take another civil service exam, and Black and Hispanic candidates in particular are more likely to be hired by the Department of Corrections (Table 3).²⁴ The DoC pipeline is far from a niche outside option: roughly 2,200 of the 6,500 entry-level Correction Officer appointments between 2014 and 2019 come from candidates who had previously sat the NYPD Police Officer exam, suggesting that outflow from the NYPD pipeline is a meaningful source of recruits for the city’s other law enforcement agency. Hence, underrepresented groups gravitate towards

²⁴To rule out mechanical effects, I exclude candidates who are hired by the NYPD within that year.

outside options during a long, uncertain process, which is consistent with interview-based evidence that minority police applicants withdraw at higher rates than White applicants (Ryan et al., 2000; Ployhart et al., 2002; Kringen and Kringen, 2014).

Table 3: Differences in Alternative Careers (Among Non-Hired NYPD Exam Passers)

	Other Exam		Hired at DoC	
Republican	-0.010*	0.001	-0.005**	-0.004**
	(0.004)	(0.004)	(0.001)	(0.002)
Non-Partisan	-0.014***	-0.006	-0.004**	-0.003*
	(0.003)	(0.004)	(0.001)	(0.001)
Black	0.084***	0.063***	0.016***	0.016***
	(0.005)	(0.005)	(0.002)	(0.002)
Hispanic	0.032***	0.022***	0.012***	0.012***
	(0.003)	(0.004)	(0.001)	(0.002)
Asian	0.039***	0.024**	0.002	0.002
	(0.006)	(0.008)	(0.002)	(0.002)
Other Race	0.049***	0.021**	0.005**	0.004
	(0.006)	(0.007)	(0.002)	(0.002)
NYPD Exam $\times$ Score Bin FE	Yes	Yes	Yes	Yes
Additional controls	No	Yes	No	Yes
Mean of DV	0.149	0.147	0.019	0.019
Adj. R ²	0.035	0.043	0.020	0.019
Num. obs.	62068	57390	60249	55759

Linear probability regressions, weighted by the posterior probability of a match between the respective data sets. Dependent Variables: Taking another civil service exam within a year of NYPD exam (Model 1,2); Being appointed as corrections officer (Model 3,4). Additional controls include gender, age, and county of residence. HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Fourth, financial resources differ systematically across exam passers. Non-hired exam passers and those taking another civil service exam cluster at the lower end of the income distribution (Figure A13), with the pattern more pronounced among non-White and Democratic exam passers (Figure A14).²⁵ Lower income may constrain candidates' ability to absorb a long, uncertain process, particularly when alternatives in the broader civil service

²⁵Income is estimated from L2's household income measure and per-capita income from the 2019 American Community Survey at the census-tract level. Because L2 income is observed in 2021, hired candidates' income partly reflects post-appointment NYPD salary; the across-group comparisons among *non-hired* candidates are therefore the cleaner test.

offer faster appointment timelines.

6.2 Selector Effects and Candidate Quality

Alternatively, these patterns could be a result of selector effects, where the NYPD itself screens candidates disproportionately at the long discretionary post-exam process, or at discretionary promotions, assignment, and exit stages that follow.

One version is selection on *objective quality*, where Republicans and Whites get hired and promoted more because of latent, policing-relevant qualities that the written exams do not capture. Several pieces of evidence cut against this. The fact that unhired underrepresented NYPD exam takers are more likely to secure positions at the Department of Corrections suggests that these candidates are clearly appointable for law enforcement work, and that the same candidates with the same measured ability end up in different agencies depending on which discretionary process they go through. Additionally, hired Republican and White officers do not have better, if anything slightly *worse*, disciplinary records on the job, as measured by CCRB complaints three years after hiring (Figure A15). While this analyses cannot account for officer assignments or civilian complaint behavior, it cuts against the idea that these officers are higher quality to begin with. Additionally, no widening of observable quality gaps appears across tenure (Figure A16), inconsistent with the prediction that minority officers diverge in quality from White officers over time. And the demographic gaps in promotions and exits persist after controlling for disciplinary records as a rough proxy for ability. Hence, it is unlikely that the observed representational disparities in the NYPD are explained by latent differences in ability or merit.

A second, and more plausible, version is *selection on the institution's own preferences*, where the agency favors candidates and officers who conform to its norms and behaviors. “Fit” here means matching the prevailing organizational culture, sharing political views, communicating in expected ways, and acting like the typical officer. This version need not require any individual gatekeeper to hold biased preferences. It can be taste-based, where

individual evaluators prefer certain candidate types. It can also be more innocuous, such as statistical discrimination where evaluators use group membership as a proxy for unobserved characteristics they cannot directly measure, or a more diffuse process in which the agency maximizes on its own organizational definition of a “good” officer (Simon, 2024). What unites these variants is a self-reinforcing logic. The agency selects on its current notion of fit, and the selected officers reinforce that notion as they accumulate. Over time, the working definition of a good officer becomes the description of the typical incumbent, and selection on fit becomes selection on the demographic and partisan composition the institution already has. The evidence on differential promotions, assignments, and involuntary attrition supports this view (Tables 2, A12 and A13 and fig. 8). Because it’s harder for officers to select into an award or out of a dismissal, these gaps point to selection by the agency rather than sorting by candidates.

A related dynamic may operate on incumbents themselves. Once inside the agency, officers may shift their political identification and behavioral norms toward the prevailing organizational culture, which produces institutional homogeneity through conformity rather than through selection (Goncalves and Tuttle, 2024). For the NYPD, however, this socialization channel does not appear to be a major driver of the partisan gap. Only about 2% of NYC voters switched between the Democratic and Republican parties in the four years preceding the 2021 L2 snapshot, and only 2.4% among the 2016–2020 exam cohort. Dropping switchers does not meaningfully alter the partisan gaps reported here (Table A8), and restricting to voters who first registered before 2018 and have not switched produces partisan gaps of similar magnitude (Table A9), which rules out strategic 2020 closed-primary registration as the driver.

Taken together, self-selection in the post-exam period together with agency selection on fit through the discretionary procedures across stages appear to be the more substantial channels.

7 Quantifying the Relative Contribution of Each Stage

What is the relative impact of each stage—attraction, selection, promotion, and retention—on the representational disparities we see? To what extent could the NYPD hypothetically mitigate these gaps by adjusting their policies at each stage?

To address these questions, I provide a simple exercise in the spirit of [Chetty et al. \(2023\)](#). This analysis estimates the potential increase in the number of Democrats and non-Whites on the NYPD payroll under scenarios where application rates, hiring rates, career progressions, and retention rates are equalized across groups one at a time. In each scenario, Democrats are assigned the rate observed for Republicans, and Blacks the rate observed for Whites. It's important to note that this simulation is not designed to predict actual outcomes of policy interventions, as it does not account for behavioral responses or general equilibrium effects that would likely occur in practice. Additionally, this exercise relies on observed differences in rates across groups that may themselves be influenced by factors not captured in the data. The analysis should therefore be interpreted as quantifying the relative importance of each selection stage using observed disparities rather than isolating the causal impact of reforming selection procedures. Utilizing the estimates of application rates, hiring rates, rank assignments, and attrition rates from Section 5, I calculate these outcomes in the following way.

Applications and Passing of First Entry Exam. Using Equation (1), I calculate the actual and the possible number of appointments from underrepresented groups u (e.g., Democrats or Blacks). I compute possible entry by replacing underrepresented groups' apply or hire rate with the corresponding rate among overrepresented groups, holding the other stage fixed

$$\text{Entry}_u = N \text{ Eligible}_u \times \text{ApplyRate}_u \times \text{HiringRate}_u \quad (1)$$

$$\text{Simulated Entry}_u = N \text{ Eligible}_u \times \text{ApplyRate}_o \times \text{HiringRate}_u \quad (2)$$

This analysis is subject to a few important caveats: First, for partisanship, I have to approximate the number of eligible individuals using information on the number of *NYC voters* in each group who satisfy the minimum application requirements at the NYPD (qualifying age, education, residency). For race, I use estimated counts of eligible citizens by group from the 2019 Census Bureau’s American Community Survey (ACS).²⁶ Additionally, due to the nature of my exam data, the variable *ApplyRate* inherently combines both the willingness to apply and the ability to pass the entry exam.

Hiring. Similarly, I calculate the possible number of appointments in the absence of disparities in hiring rates by assuming that each applicant from underrepresented groups u is hired at the same rate as aspirants from overrepresented demographics o .

$$\text{Simulated Entry}_u = N \text{ Eligible}_u \times \text{ApplyRate}_u \times \text{HiringRate}_o \quad (3)$$

Table 4: Equalization Exercise - Entry

	N Eligible	N Applicants	Apply Rate	Sim. Apply Rate	Hire Rate	Sim. Hire Rate	Actual Entry	Sim. Entry Apply	Sim. Entry Hire
Democrats	1073362	6572	0.006	0.014	0.104	0.185	683	1596	1213
Blacks (L2)	298594	2062	0.007	0.009	0.104	0.132	215	268	273
Blacks (Census)	354659	2062	0.006	0.005	0.104	0.132	215	198	273

The simulated rates for Democrats (Blacks) are taken from Republican (White) applicants. $N \text{ Applicants}$ is the total number of applications observed during the sample window divided by the years observed (13,400 per year), times the share of the respective demographic from Figure 4. *ApplyRate* is estimated as $N \text{ Applicants} / N \text{ Eligible}$. *HireRate* is estimated from predicted hiring probabilities based on Table 1, Column 3.

Table 4 shows the results of this exercise. If application and hiring rates remain unchanged, the NYPD is expected to add an average of 683 Democrats and 215 Black aspirants annually. For partisanship, the application stage is the dominant lever. Equalizing application rates more than doubles Democratic recruitment to 1,596 (+134%), and more than the 1,213 Democrats (+78%) added by equalizing hiring rates. This is because the Republican-to-Democrat rate ratio is bigger at apply (2.3×) than at hire (1.8×). For race, hire-rate

²⁶<https://www.census.gov/programs-surveys/acs/microdata.html>

equalization is the more reliable lever. Whether application-rate equalization helps depends on how the eligible pool is measured. Under the L2-based estimate, it increases Black recruits by 25% (close to the 27% from equalizing hire rates), while under the more inclusive ACS Census estimate it slightly *worsens* representation because Black applicants already apply at higher rates relative to their eligible pool than Whites. Hire-rate equalization yields a 27% increase across both eligible-pool measures.

Retention. Following a similar logic, I also estimate how many additional underrepresented officers the NYPD could retain by eliminating differences in career progression and attrition rates, and again compare these quantities to the actual predicted number of retained officers. For this, I use the predicted rate of officers from each group to be assigned to rank r and retained, conditional on officers' years of service (binned) and age.

$$\text{Retention}_u = N \text{ Officers}_u \times \sum_r \text{Rank}_{r,u} \times \text{RetentionRate}_{r,u} \quad (4)$$

$$\text{Simulated Retention}_u = N \text{ Officers}_u \times \sum_r \text{Rank}_{r,o} \times \text{RetentionRate}_{r,u} \quad (5)$$

$$\text{Simulated Retention}_u = N \text{ Officers}_u \times \sum_r \text{Rank}_{r,u} \times \text{RetentionRate}_{r,o} \quad (6)$$

In contrast, equalizing retention rates and rank distributions has only marginal effects on the composition of the force. Even if the NYPD were to retain Democrats and Blacks at the same rates as Republicans and Whites, the net change would amount to fewer than 20 additional officers per year (Table 5). This is primarily because overall retention rates are already high across groups (approximately 95%), leaving little room for improvement.

Overall, this exercise illustrates that the most effective lever for enhancing the representativeness of one of America's largest law enforcement agencies is equalizing hiring and, for partisanship, application rates across under- and overrepresented groups.

Table 5: Equalization Exercise - Retention

	N Officers	Actual Retention	Sim. Retention Ranks	Sim. Retention Rate
Democrat	11996	11446	11441	11454
Black	3427	3287	3283	3265

The CF rates for Democrats (Blacks) are taken from Republican (White) officers. *N Officers* is the average number of uniformed employees on the payroll across years in my sample by the respective demographic.

8 Conclusion

Civil service systems were built to make government employment turn on merit rather than connections, and the debate over whether they succeed has predominantly centered on the written entry exam. This article instead asks where within a civil service system representational gaps arise. Decomposing the pipeline of the NYPD, it shows how multiple stages of self-selection, recruitment, promotion, and attrition combine to produce bureaucratic composition, with the largest gaps emerging at the discretionary processes that follow the formalized civil service exams.

These findings reframe a long-standing debate over civil service reform and meritocracy in public hiring. Both reformers and critics of civil service have concentrated their attention on the written entry exam. Reformers see it as the central diversification lever (Van Riper et al., 1963; Hoogenboom, 1968), while critics see it as the principal site at which ostensibly neutral rules reproduce existing inequalities (Portillo et al., 2020; Kuipers and Sahn, 2023; Moreira and Pérez, 2025; Kuipers, 2025). The pipeline decomposition presented here suggests that this shared focus has treated civil service as a binary system, missing how meritocratic rules and the discretion they leave intact interact across the full bureaucratic career. Reform efforts that target a single formal procedure, however carefully designed, are unlikely to close gaps that are produced cumulatively across many discretionary decisions over a career.

The findings also reframe how we think about representative bureaucracy. A growing literature studies the consequences of bureaucrats’ demographic and partisan identity for

service provision (Bradbury and Kellough, 2011; Ba et al., 2021, 2025; Donahue, 2023), but the findings are often inconclusive and the conditions under which representation translates into more equitable outputs remain unsettled (Meier, 2018). Understanding the *origins* of representation is critical to reconcile this ambiguity, because the same level of demographic composition can produce different outcomes depending on how that composition is achieved. At the NYPD, the discretionary procedures that drive most of the variation in composition do not appear to select on measurable qualifications. Democratic and non-White officers who successfully navigate these procedures may instead be those who most closely align with the prevailing organizational culture. Even those who are not aligned at entry may face strong incentives to conform once inside, since advancement in a predominantly White and Republican environment continues to depend on discretionary judgments by superiors who share that culture. Simply achieving demographic diversity may therefore be insufficient if both the selection processes that admit minority officers and the institutional incentives they face favor those aligned with established norms. To theorize the effects of demographic shifts in police forces, and bureaucracy more broadly, we need to incorporate the mechanisms through which representation arises and is sustained (Ashworth et al., 2024).

Yet, important issues and open questions remain. Although the large administrative data from NYC allows for a rich picture of bureaucratic selection, the analysis is restricted to only one primary US jurisdiction, and it remains to be seen how the results generalize to other city governments. Two features of the case nonetheless suggest that the broader insight may travel. First, the institutional structure that produces the asymmetry is shared across most large US civil service agencies. Formalized exam-based procedures coexist with discretionary vetting, promotion, assignment, and termination decisions in police, fire, corrections, and other local bureaucracies across the country (Riccucci and Riccardelli, 2015; Anzia and Trounstein, 2025). Second, NYC is a hard case for substantial political sorting. The NYPD’s civil service rules are well-codified and its hiring pipeline includes substantial formalization. If sorting nonetheless concentrates in the discretionary procedures within this institutional

setting, the same pattern may operate elsewhere and even more sharply where formal rules are thinner.

Additionally, the evidence is descriptive and cannot identify the exact mechanisms behind the documented disparities in bureaucrats’ partisanship and race. Triangulating across the available sources already narrows the field, but measuring the surviving explanations directly would require different data. For example, future research could use records of why and when candidates leave the hiring process to separate voluntary withdrawal from agency rejection. Data on which investigators screen which candidates would support comparisons across evaluators facing similar applicants. And surveys of candidates and officers would provide more details on motivation and intent.

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Appendix: Supporting Information for *The Limits of Merit*

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A Merging Procedures and Quality

In this section, I describe the different merging procedures and the quality of the resulting matches.

A.1 Merge Payroll to L2

As described in the main text, I start with a roster of the six largest agencies in NYC. I identify individual employees by unique combinations of appointment date, last name, first name, and middle initial. I then use a probabilistic record linkage algorithm to match demographic information for all employees from the L2 voter file (Enamorado et al., 2019), retaining all matches with a posterior match probability of at least 0.7. For first and last names, I allow for partial string distance matches using the Jaro-Winkler distance methods, while for the middle initial I enforce an exact match.

Since L2 does not include information on appointment dates and I need to rely on name information for these matches only, one might be concerned that this introduces a large amount of duplicates in the list of unique employees. However, note that the combination of first name, last name, and middle initial (i.e., excluding appointment date) already uniquely identifies 95% of all bureaucrats. Similarly, 87% of all voter names are unique in the L2 data. Importantly, these shares are very similar across party and race groups: 89% for Democrats, 95% for Republicans, 95% for Independents, 91% for Whites, 91% for Blacks, 85% for Asians, 74% for Hispanics and 92% for other races.

Table A1 shows that I am able to identify more than 80% of bureaucrats across all agencies, and Figure A1 indicates that the median posterior probability for these matches is more than 0.95. Additionally, Table A1 shows that the true match rate is estimated to be at least 69% across the agencies, and with low false discovery rates (FDR) and high false negative rates (FNR) the algorithm clearly errs on the side of not identifying a bureaucrat in the voter file rather than matching the wrong voter to a bureaucrat. More importantly, the FDR is relatively similar across race and party groups, albeit somewhat lower for Republicans, Whites and male bureaucrats across the agencies.¹ It’s important to note that I weight by the posterior probability of matches in all analyses to quantify the uncertainty inherent in my merge procedures, and to calibrate and account for the amount of false positives and false negatives across demographic groups in my data.

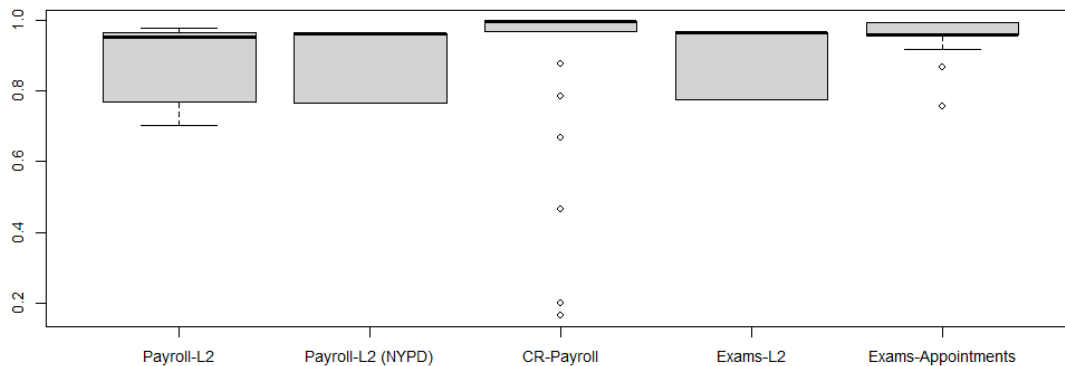
¹Unfortunately, the FNR is computationally impossible to calculate by race or partisanship, since this would require retaining matches for *all* pairs across the two data sets, i.e. $N_{bureaucrats} \times N_{voters} \geq 168,577,197,264$.

Table A1: Merging of NYC Payroll (2014-2021) to 2021 L2 Voter File (N=7,940,144)

	NYPD (N=91,975)	Sanitation (N=32,468)	FDNY (N=28,016)	Social Services (N=22,386)	Corrections (N=21,231)	Education (N=189,772)
Number of matches	80,661 (88%)	26,618 (82%)	26,172 (93%)	18,909 (85%)	18,345 (87%)	85,798 (45%)
True match rate ^a	78%	73%	87%	69%	74%	
False negative rate (FNR) ^b	94%	95%	96%	92%	93%	
False discovery rate (FDR) ^c	11%	11%	7%	18%	15%	
FDR by groups						
Non-Partisan	11%	11%	7%	20%	15%	
Democrat	12%	12%	8%	17%	14%	
Republican	9%	11%	6%	21%	15%	
White	9%	10%	6%	18%	14%	
Hispanic	13%	13%	9%	20%	17%	
Black	11%	11%	8%	16%	13%	
Other	14%	14%	10%	23%	16%	
Race						
Asian	15%	16%	11%	22%	18%	
Female	12%	12%	9%	18%	14%	
Male	10%	11%	7%	18%	15%	

^a Share of matches weighted by their posterior probability of a match; ^b Probability of wrongfully declaring a non-match given the threshold; ^c Probability of wrongfully declaring a match given the threshold

Figure A1: Boxplots of Posterior Probabilities of Correct Matches across Matching Procedures



A.2 One-to-Many Matches

An issue when matching administrative records that lack a large number of unique identifiers and matching variables is the possibility of one-to-many matches, i.e., several agency employees match to more than one voter. As noted early, names in both the bureaucrat list and voter files are predominantly unique. As a result, Table A2 shows that the algorithm achieves a very high rate of one-to-one matches, with the share of unique matches being above 90% across agencies (with the exception of the Department of Education). Further inspecting the remaining one-to-many matches suggests that these are due to a re-appointment of employees (e.g., a traffic enforcement agent becomes appointed as a police officer). Since these types of duplicates are warranted, I maintain one-to-many matches without further adjustments.

Table A2: Share of One-to-One Matches: Payroll-L2

	NYPD	Sanitation	FDNY	Social Services	Corrections	Education
Number of matches	80,661 (88%)	26,618 (82%)	26,172 (93%)	18,909 (85%)	18,345 (87%)	85,798 (45%)
Share of one-to-one matches	94.5%	92.8%	92%	96.9%	97.8%	80.1%

Similar levels and reasoning applies to one-to-many matches for the additional merges of NYPD datasets (Table A3). Multiple city records can match the same employee on the payroll, e.g., if the same employee receives multiple promotions or retires for different

positions. Similarly, the same voter may take the entry or promotion exam several times, thus leading to a small share of duplicates in the exam-L2 merges. Exceptions are the exam-appointment and exam-promotion match, where the same exam-taker should only be hired/promoted for the same position once. I account for these one-to-many matches in my matching procedures, as outlined in sections A.5 and A.6.

Table A3: Share of One-to-One Matches: Other NYPD matches

	City Record- Payroll	Entry Exams-L2	Entry Exams- Appointments	Promo Exams-L2	Promo Exams- Promotions
Share of one-to-one matches	90.5%	88.9%	64.8%	85%	98%

While these figures alleviate concerns about duplicates in voter names, one may still be concerned that matched names differ in their homogeneity of party and race. That is, if, for example, names in the bureaucrat list matched to Republicans have a larger share of Democrats with the identical names in the voter files than Democratic bureaucrat names have Republicans in the voter file, one-to-one matches could lead to bias and an overestimation of Republicans in the police. To assess this, I estimate group homogeneity in the voter file for all names that I classify as matches in the bureaucrat list. As Table A4 shows, differential name purity is of little concern. If anything, Republican and White names in the officer list have *higher* name purity than Democratic and Black names. Hence, it is unlikely that I artificially inflate the number of Republicans and Whites in the matched set due to mismeasured partisanship or race among their names.

Table A4: Name Purity of Matches: Payroll-L2

	Repub- lican	Demo- crat	Non- Partisan	Asian	Black	Hispanic	Other Race	White
Purity by Full Names	93%	92%	90%	99.9%	96.3%	100%	99.9%	98.8%

Purity measures are maximum proportions of (partisan/race) groups by names, averaged by matched party or race.

A.3 Merge City Records with NYPD Payroll

For my analysis of the selection dynamics at the NYPD, I collected various additional data sets. Particularly, I obtained information on (1) career milestones, including appointments,

promotions, and retirements from official records published in the daily City Record newspaper since 2014²; (2) unit assignment, awards, and arrest records for a cross-section of about 33,000 active officers (as of October 2021) from NYPD’s official officer profiles³; (3) civil service exams, both for entry and promotions.⁴ I then match these different data sets to the roster of NYPD bureaucrats. The following sections describe these various matches in more detail. Table A5 illustrates the number of successful matches across data sets, and Figures A2 and A3 illustrate the high posterior probability of a match across the procedures together with slight differences across demographic groups.

I link the 65,856 City newspaper records on appointments, promotions, demotions, retirements, resignations, dismissals, and terminations to the roster of NYPD bureaucrats in the following way: To maximize overlap, I restrict the city records to those with effective date between January 1, 2014, and July 1, 2021. I then match on first name, last name, and middle initial, and retain matches with a minimum posterior of 0.7. I do not deduplicate matches returned by the algorithm. Instead, I deal with multiple city records that match to the same payroll names in the following way:

- In cases where there is one maximum posterior, we keep the maximum posterior match (equivalent what fastLink enforces as default)
- For cases where there is no unique maximum posterior probability, I use additional information in the data.
- For *appointments*, I identify exact matches using the appointment date provided in both data sets. If there is more than one exact match, I use the maximum posterior probability for these duplicates again and if these maxima are not unique, I retain the payroll entry with the earliest appointment. If there are no exact matches using the appointment date, I retain those with the smallest difference between the appointment date in the payroll and the effective date in the city records. If there is more than one such time distance match, I use the maximum posterior match probability for these duplicates again and if these maximum posteriors are not unique, I retain the payroll entry with the earliest appointment.
- For *attrition*, I again follow the default in fastLink and use the maximum posterior to find the best match in cases of multiple matches based on name. If there are more than one such maximum posterior match, I use the smallest difference between the fiscal year at the date of attrition and the last fiscal year recorded in the payroll. If this still does not give a unique match, I use the employee with the earliest appointment date, i.e. the person who has been on the force the longest before retirement.
- For *promotions*, I use information on changes in job titles on the payroll to adjudicate between multiple maximum-posterior matches. I first retain the maximum posterior among all promotion matches. If there is more than one possible match, I use the year

²<https://www.nyc.gov/site/dcas/about/cityrecord-editions.page>

³<https://nypdonline.org/link/2>

⁴I carefully combined both active and terminated civil service lists as of July 2022 from <https://open.data.cityofnewyork.us/>.

of promotion that matches the promotion in the city records. If this is not the case, I use the minimum difference in the year of the title change on the payroll and the year of promotion in the city records. For the remaining duplicates, I use the employee who was appointed first. For the remaining 4 duplicates, I finally use the first observation in the data.

Overall, I am able to successfully match 65,184 (99%) of 65,856 city records.

A.4 Merge NYPD Payroll to NYPD Online Officer Profiles

I used another iterative process to match the roster of active officers from NYPD online to my payroll roster. I first merge these records based on exact matches using the appointment date, the first name, the last name, and the middle initial. All payroll records I fail to match exactly, I then also match based on a fuzzy name match and extensive manual checks by research assistants.

A.5 Merge Entry Exams to Appointments and L2 Voter File

To match entry-level exams to the roster of NYPD personnel, I collect the information on 96,883 NYC civil service exams for police officer and traffic enforcement agents. I rely on both the active and terminated civil service lists as of July 2022.⁵ Following a probabilistic record linkage based on the first name, last name and middle initial, I first retain all matches with a minimum posterior probability of 0.5. I then rely on the following procedure to deduplicate matches of the same exam: First, I restrict matches by enforcing that the police officer title of the exam corresponds to the title of the employee’s appointment. I further deduplicate matches by requiring that the date of the exam needs to be before the date of appointment. I further filter duplicates by retaining the maximum posterior probability match by exam. For the remaining duplicates, I use the minimum distance between the appointment date and the exam date. I also account for duplicates in the appointments for the same exam-taker, e.g., because some hired NYPD employees took the exam more than once before getting hired. I again rely on the maximum posterior match and the exam date relative to the appointment date to identify the successful exam for these individuals. Finally, I also match the entry exams to the L2 voter file. For computational reasons, I first estimate the EM step of the fastLink algorithm on a random 50% sample of the exam passers and voters, respectively. I then apply the parameter estimates to all exam-L2 combinations, and block on the first letter of the last name for computational tractability. Also, I restrict the eligible voters to those younger than 45 (in 2020) to account for the age restrictions in the NYPD hiring process.

I am able to match 10% of all exam passers to NYPD appointments and identify 82% of all appointments between 2014 and 2021 among exam passers. The non-matched appointments are mostly from earlier years (e.g., 63% appointments in 2014 remain unmatched to exams) due to truncation and delayed temporal overlap between exams and appointments.

⁵https://data.cityofnewyork.us/City-Government/Civil-Service-List-Active-/vx8i-nprf/about_data; https://data.cityofnewyork.us/City-Government/Civil-Service-List-Terminated-/qu8g-sxqf/about_data

A.6 Merge Promotion Exams to Awarded Promotions and L2 Voter File

Individuals who seek promotions to the ranks of Sergeant, Lieutenant, and Captain need to successfully pass additional civil service exams at the NYPD. I match promotion exams to the promotions recorded in the city records probabilistically based on employee's first name, last name and middle initial. After retaining matches with a minimum posterior match probability of 0.5, I again ensure that exam titles correspond to the titles of the matched promotion records. Similar to the entry exams, I also require that the date of the promotion exam is before the date of the promotion, and use the minimum time difference between the exam date and the promotion date for the remaining duplicates. To ensure that I record only one promotion for a successful exam-taker (i.e., rule out duplicates in promotions for the same exam passers), I follow a similar procedure to the entry exams: I again rely on the maximum posterior match and the exam date relative to the appointment date to identify the successful exam for these individuals. Finally, I also match the promotion exams to the L2 voter file, following exactly the same procedure as for the payroll-L2 merging described above.

Table A5: Merging of NYPD Data Sets

	L2 Voter File, 2020 (N=7,940,144)	City Records, 2014-2021 (N=65,856)	Active Offi- cer Profiles (N=33,072)	Other Civil Service Exams (N=195,308)
NYPD Payroll*, FY2014-2021 (N=91,975)	80,661 (88%, .)	65,184 (., 99%)	32,632 (., 99%)	
Officer En- try Exams, 2014-2021 (N=99,159)	78,894 (80%, .)	11,763 (12%, 81% ^a)		19,651 (20%,.)
Promotion Ex- ams, 2012-2021 (N=5,725)	5,226 (91%,.)	3,501 (61%,.)		

Percentages of matched observations in parentheses (row percentages, column percentages); * includes both uniformed and civilian personnel; ^a among NYPD POLICE OFFICER and POLICE CADET appointments only (N=14,528)

Figure A2: Average Posterior Probabilities of Correct Matches By Groups across Matching Procedures - Payroll Data

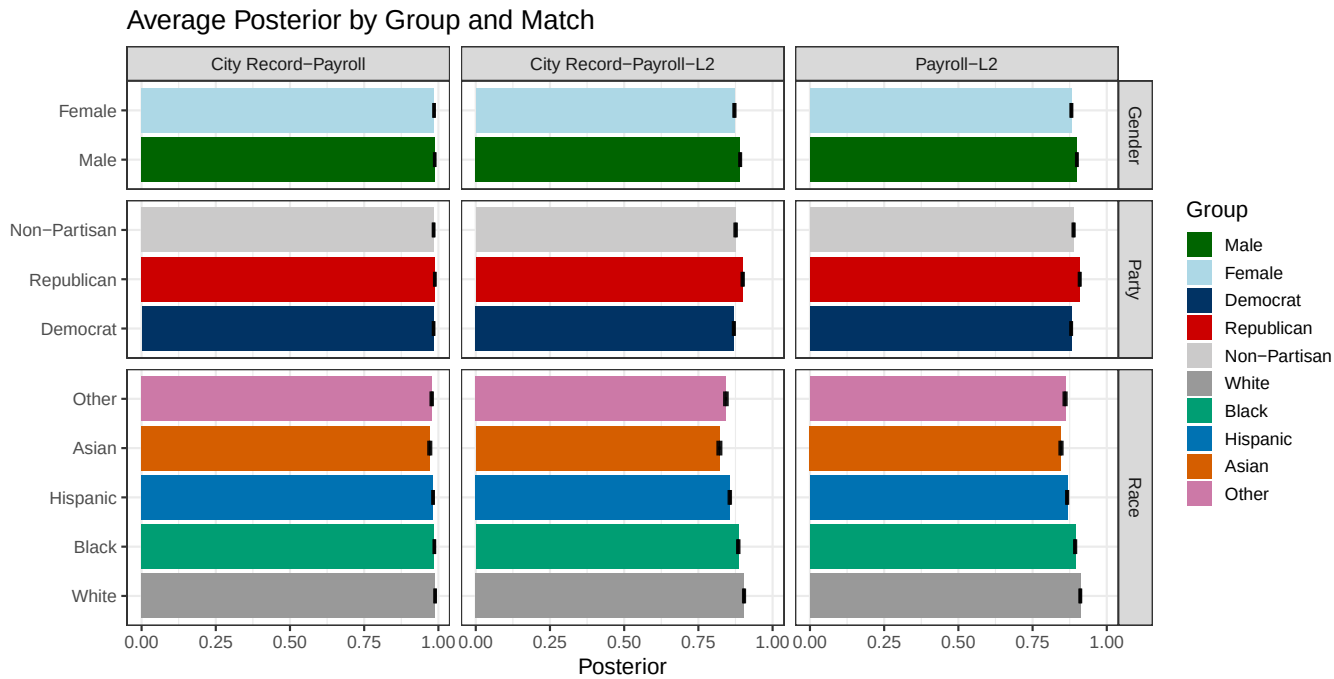
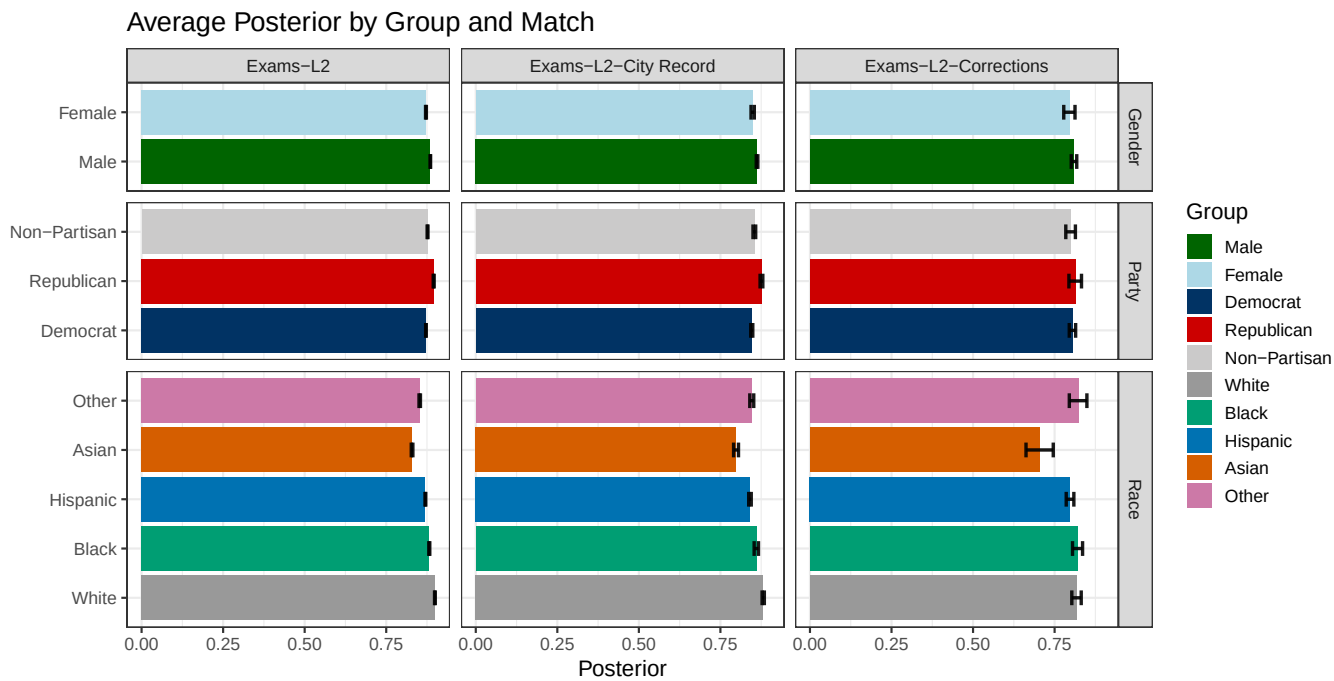


Figure A3: Average Posterior Probabilities of Correct Matches By Groups across Matching Procedures - Exam Data

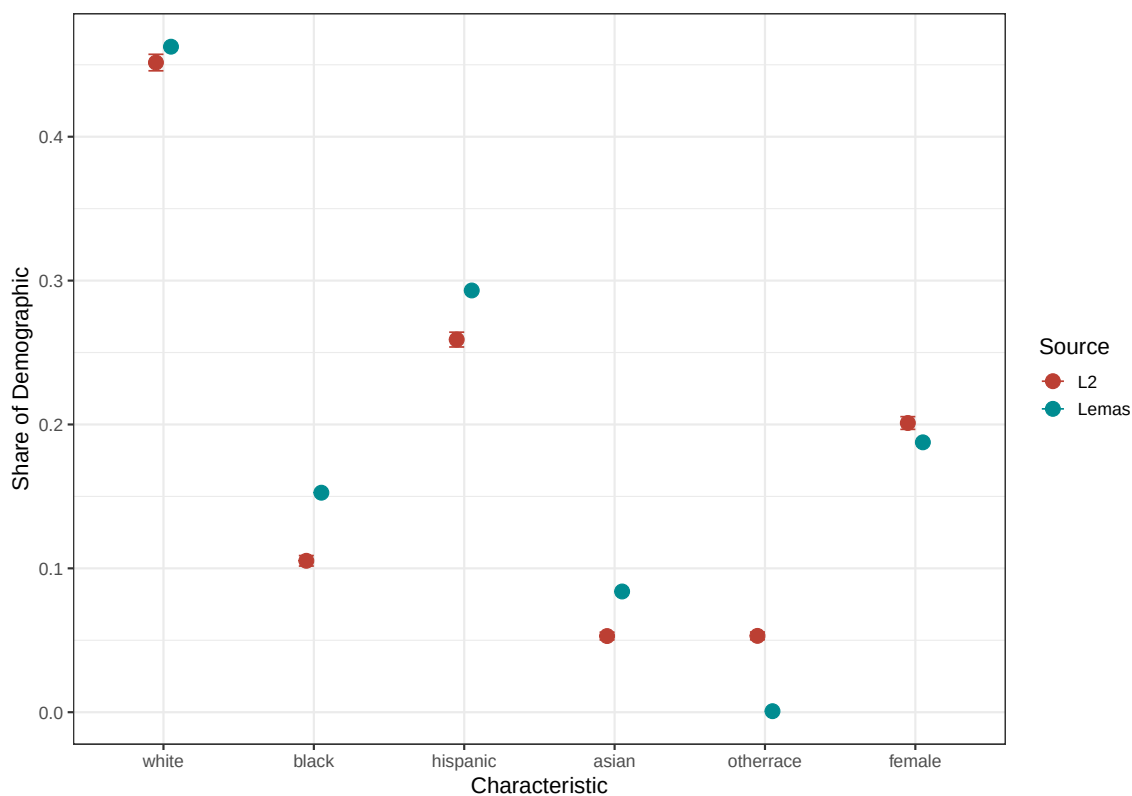


A.7 Measurement Error in L2 Race/Ethnicity

I assess the validity of the L2 race coding in two ways.

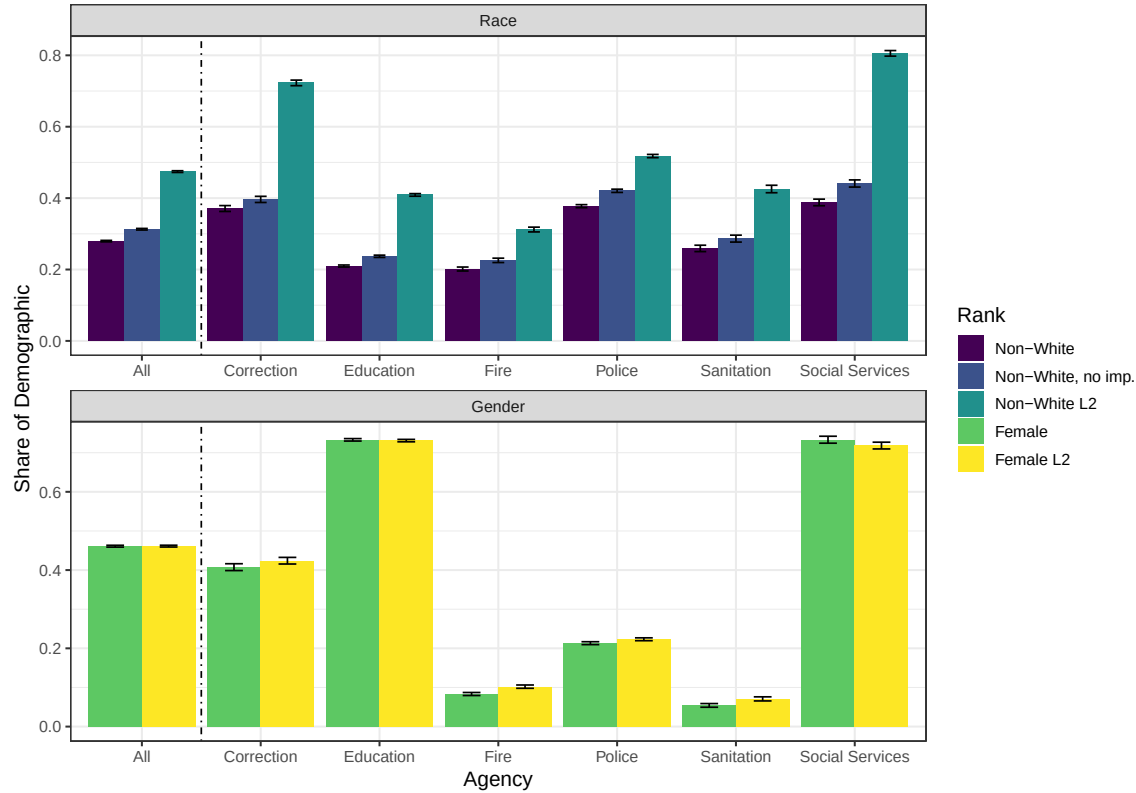
First, the LEMAS survey asks police agencies to report the number of sworn full time employees across different demographic groups (particularly, race and gender). I draw on the answers of the NYPD in the 2020 LEMAS survey to assess the validity of the L2 information on race and gender. Figure A4 depicts the proportion of officers in each racial/ethnic category and by gender as measured by L2 vs. LEMAS. To ensure optimal overlap in the populations used in both data sets, I restrict the payroll to only sworn personnel and weight estimates by the posterior probability of a match. As the figure shows, L2 slightly underrepresents the proportion of White officers, and more so, the share of other minorities. This discrepancy largely stems from the overrepresentation of the “other/unknown” category, which makes up 5% in the L2 data, but essentially 0% in the LEMAS data. The L2 data also slightly overrepresents female officers compared to the LEMAS data.

Figure A4: Comparison of 2020 LEMAS and L2 Measures of Officer Race



Second, I rely on predictions of individual-level ethnicity of NYC employees using Imai and Khanna (2016). Figure A5 compares the estimated share of non-White employees across agencies by types of measures. Evidently, the share of non-White bureaucrats is estimated to be substantially higher with L2’s race measures than alternative predictions across agencies. Taken together, these tests suggest that I slightly *overestimate* the degree of representation for minority groups in NYC’s bureaucracy, i.e., the representational gaps may be even starker than I estimate.

Figure A5: Comparison of Bayesian Name Predictions and L2 Measures of Bureaucrat Race



Note: “Non-White” and “Non-White, no imp.” use Bayesian surname-based race predictions from the `wru` R package; “no imp.” restricts to individuals whose surname matched directly in the Census surname list. “Non-White L2” uses L2’s proprietary race imputation based on name and census-block demographics.

B Additional Tables

B.1 Entry Exams

Table A6: Differences in Exam Scores across Demographics

	Model 1	Model 2	Model 3	Model 4
Republican	0.04 (0.08)	-0.01 (0.07)	-0.05 (0.07)	0.36*** (0.07)
Non-Partisan	0.00 (0.07)	-0.07 (0.06)	-0.06 (0.06)	0.22*** (0.06)
Black	-1.70*** (0.09)	-1.78*** (0.09)	-1.77*** (0.09)	-2.83*** (0.09)
Hispanic	-1.15*** (0.07)	-1.28*** (0.07)	-1.25*** (0.07)	-2.28*** (0.07)
Asian	0.26* (0.12)	0.15 (0.12)	0.20 (0.12)	-1.19*** (0.12)
Other Race	-1.83*** (0.11)	-1.75*** (0.11)	-1.71*** (0.11)	-3.02*** (0.11)
Veteran Credit		6.19*** (0.12)	6.21*** (0.12)	6.27*** (0.12)
Parent/Sibling Legacy Credit			9.47*** (0.41)	9.56*** (0.40)
Residency Credit				3.76*** (0.06)
Exam FE	Yes	Yes	Yes	Yes
Mean of DV	88.82	88.82	88.82	88.82
Adj. R ²	0.03	0.07	0.08	0.13
Num. obs.	69406	69406	69406	69406

Regressions weighted by posterior probability of a match between exam data and voter file, and between the exams and appointments. Dependent variable: Exam score in entry-level exam (police officer exams only). Veteran credits are awarded for veterans and disabled veterans. Parent and sibling legacy credits are additional credits on the exam score to candidates who lost a parent/sibling as a result of 911. Residency credits are given on exams to candidates who maintain a continuous period of residency in NYC. HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A7: Differences in Hiring – Robustness to Functional Form of the Exam-Score Control

	Model 1	Model 2	Model 3
Republican	0.08*** (0.00)	0.08*** (0.00)	0.08*** (0.00)
Non-Partisan	0.03*** (0.00)	0.03*** (0.00)	0.03*** (0.00)
Black	−0.03*** (0.00)	−0.03*** (0.00)	−0.03*** (0.00)
Hispanic	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Asian	0.02** (0.01)	0.01* (0.01)	0.01* (0.01)
Other Race	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)
Exam Score Q2 (within-exam quintile)	0.01* (0.00)		
Exam Score Q3 (within-exam quintile)	0.02*** (0.00)		
Exam Score Q4 (within-exam quintile)	0.02*** (0.00)		
Exam Score Q5 (within-exam quintile)	0.02*** (0.00)		
Within-exam rank: top 10–25%		0.01** (0.00)	
Within-exam rank: top 5–10%		0.01 (0.01)	
Within-exam rank: top 1–5%		−0.00 (0.01)	
Within-exam rank: top 1%		−0.07*** (0.01)	
Within-exam percentile rank (0–1)			0.02*** (0.00)
NYPD Exam Cohort FE	Yes	Yes	Yes
Score control	Within-exam quintile	Within-exam rank bracket	Continuous percentile
Mean of DV	0.13	0.13	0.13
Adj. R ²	0.04	0.04	0.04
Num. obs.	69406	69406	69406

Linear probability regressions, weighted by the posterior probability of a match between exam data and voter file, and between exam data and appointments. Column 1 reproduces the main Table 1 specification using within-exam score quintiles. Column 2 replaces them with mutually exclusive within-exam rank brackets (reference: bottom 75%). Column 3 uses a continuous within-exam percentile rank (0 = bottom, 1 = top). Reference categories: Democrat (party) and White (race). HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A8: Differences in Hiring By Exam Passer Characteristics – Party-Switcher Robustness (2016–2020)

	Model 1	Model 2	Model 3	Model 4
Republican	0.06*** (0.01)	0.06*** (0.01)	0.05*** (0.01)	0.04*** (0.01)
Non-Partisan	0.02*** (0.00)	0.02*** (0.00)	0.02** (0.00)	0.01** (0.00)
Black	−0.03*** (0.01)	−0.03*** (0.01)	−0.01 (0.01)	−0.01* (0.01)
Hispanic	−0.00 (0.00)	−0.00 (0.00)	0.01 (0.01)	0.01 (0.01)
Asian	0.00 (0.01)	−0.00 (0.01)	0.01 (0.01)	0.01 (0.01)
Other Race	−0.01 (0.01)	−0.01 (0.01)	0.00 (0.01)	−0.00 (0.01)
Male			0.03*** (0.00)	0.03*** (0.00)
County: Kings			−0.01* (0.01)	−0.01* (0.01)
County: Nassau			0.03*** (0.01)	0.03*** (0.01)
County: New York			−0.01 (0.01)	−0.01 (0.01)
County: Queens			−0.00 (0.01)	0.00 (0.01)
County: Richmond			−0.00 (0.01)	−0.00 (0.01)
County: Rockland			0.04** (0.02)	0.04** (0.02)
County: Suffolk			0.01 (0.01)	0.01 (0.01)
County: Westchester			0.01 (0.01)	0.01 (0.01)
Age			−0.00*** (0.00)	−0.00*** (0.00)
NYPD Exam×Score Bin FE	Yes	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes
Drop Party Switchers	No	Yes	No	Yes
Mean of DV	0.13	0.13	0.13	0.13
Adj. R ²	0.03	0.03	0.04	0.04
Num. obs.	31035	30106	28608	27755

Linear probability regressions on the 2016–2020 list-establishment subsample, weighted by the posterior probability of a match between exam data and voter file. "Drop Party Switchers" columns restrict to candidates with no recorded party change in the L2 file. Reference categories: Democrat (party), White (race), Female (gender), and Bronx (county). HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A9: Differences in Hiring By Exam Passer Characteristics – Party-Switcher Robustness (2016–2020 + Pre-2018 Registration)

	Model 1	Model 2	Model 3	Model 4
Republican	0.06*** (0.01)	0.05*** (0.01)	0.04*** (0.01)	0.03*** (0.01)
Non-Partisan	0.02*** (0.01)	0.02** (0.01)	0.01 (0.01)	0.01 (0.01)
Black	−0.03*** (0.01)	−0.04*** (0.01)	−0.01 (0.01)	−0.01* (0.01)
Hispanic	−0.00 (0.01)	−0.01 (0.01)	0.01 (0.01)	0.01 (0.01)
Asian	−0.00 (0.01)	−0.00 (0.01)	0.00 (0.01)	−0.00 (0.01)
Other Race	−0.01 (0.01)	−0.02* (0.01)	−0.00 (0.01)	−0.01 (0.01)
Male			0.03*** (0.00)	0.03*** (0.00)
County: Kings			−0.02* (0.01)	−0.02* (0.01)
County: Nassau			0.04*** (0.01)	0.04*** (0.01)
County: New York			−0.01 (0.01)	−0.02 (0.01)
County: Queens			0.00 (0.01)	0.00 (0.01)
County: Richmond			−0.00 (0.01)	−0.00 (0.01)
County: Rockland			0.03 (0.02)	0.03 (0.02)
County: Suffolk			0.02 (0.01)	0.02 (0.01)
County: Westchester			0.01 (0.01)	0.01 (0.01)
Age			−0.00*** (0.00)	−0.00*** (0.00)
NYPD Exam×Score Bin FE	Yes	Yes	Yes	Yes
Additional controls	No	No	Yes	Yes
Drop Party Switchers	No	Yes	No	Yes
Mean of DV	0.13	0.12	0.13	0.13
Adj. R ²	0.03	0.03	0.04	0.04
Num. obs.	23196	22451	21460	20768

Linear probability regressions on the 2016–2020 list-establishment subsample, further restricted to candidates whose L2 official registration date is before 2018-01-01 (or missing), weighted by the posterior probability of a match between exam data and voter file. "Drop Party Switchers" columns additionally restrict to candidates with no recorded party change in the L2 file. Reference categories: Democrat (party), White (race), Female (gender), and Bronx (county). HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

B.2 Career Advancement and Promotions

Table A10: Difference in Probability of Taking a Promotion Exam (2014-2021)

	Model 1	Model 2	Model 3
Republican	−0.00 (0.00)		0.00 (0.00)
Non-Partisan	0.00 (0.00)		−0.00 (0.00)
Black		−0.00 (0.00)	−0.00 (0.00)
Hispanic		0.00 (0.00)	0.00 (0.00)
Asian		0.05*** (0.01)	0.05*** (0.01)
Other Race		0.04*** (0.01)	0.04*** (0.01)
Cohort FE	Yes	Yes	Yes
Mean of DV	0.09	0.09	0.09
Adj. R ²	0.12	0.12	0.12
Num. obs.	49558	48521	45990

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table A11: Difference in Probability of Receiving Promotion After Promotion Exam (2014-2021)

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Republican	0.03 (0.01)	0.01 (0.01)			0.01 (0.01)	0.01 (0.01)
Non-Partisan	0.04* (0.02)	0.01 (0.01)			0.01 (0.01)	0.01 (0.01)
Black			-0.05* (0.02)	0.00 (0.02)	0.01 (0.02)	0.01 (0.02)
Hispanic			-0.02 (0.02)	0.00 (0.01)	0.01 (0.01)	0.01 (0.01)
Asian			-0.02 (0.03)	-0.03 (0.02)	-0.02 (0.02)	-0.02 (0.02)
Other Race			0.03 (0.03)	0.01 (0.02)	0.02 (0.02)	0.02 (0.02)
Examscore (80-90)		0.57*** (0.01)		0.57*** (0.01)	0.57*** (0.01)	
Examscore (90-100)		0.73*** (0.01)		0.73*** (0.02)	0.73*** (0.02)	
Exam FE	Yes	Yes	Yes	Yes	Yes	Yes
Exam $\times$ Score Bin FE	No	No	No	No	No	Yes
Mean of DV	0.61	0.61	0.60	0.60	0.61	0.61
Adj. R ²	0.14	0.51	0.14	0.51	0.51	0.51
Num. obs.	4963	4963	4790	4790	4549	4549

Linear probability regressions, weighted by the posterior probability of a match between promotion exams and voter file. Level of observation: Promotion exam passer. Outcome: Dummy for whether the exam passer actually received a promotion between 2014 and 2021. HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table A12: Differences in (Log) Number of Awards by Officer Characteristics

	Model 1	Model 2	Model 3
Republican	0.20*** (0.01)		0.13*** (0.01)
Non-Partisan	0.08*** (0.01)		0.04** (0.01)
Black		-0.27*** (0.02)	-0.21*** (0.02)
Hispanic		-0.16*** (0.01)	-0.12*** (0.01)
Asian		-0.26*** (0.02)	-0.24*** (0.02)
Other Race		-0.14*** (0.02)	-0.12*** (0.02)
Cohort FE	Yes	Yes	Yes
Mean of DV	0.92	0.91	0.91
Adj. R ²	0.21	0.21	0.22
Num. obs.	28422	27609	26222

Regressions weighted by posterior probability of a match between payroll and voter file. The information on awards received is only available for active officers (here as of October 2021). Level of observation: Employee. Outcome: Log number of departmental awards since their appointment date at NYPD. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table A13: Differences in Command Assignments for Active Officers (as of 10/20/2021)

	All Elite	Terrorism	Drugs	Special Forces
Republican	0.03*** (0.00)	0.00 (0.00)	0.00 (0.00)	0.02*** (0.00)
Non-Partisan	0.02*** (0.00)	0.00 (0.00)	0.00 (0.00)	0.01*** (0.00)
Black	-0.03*** (0.01)	-0.01** (0.00)	-0.00 (0.00)	-0.01** (0.00)
Hispanic	-0.01 (0.00)	-0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)
Asian	-0.03*** (0.01)	-0.00 (0.01)	-0.01*** (0.00)	-0.01** (0.00)
Other Race	-0.02** (0.01)	-0.00 (0.01)	-0.00 (0.00)	-0.02*** (0.00)
Cohort FE	Yes	Yes	Yes	Yes
Mean of DV	0.10	0.04	0.02	0.04
Adj. R ²	0.05	0.02	0.01	0.02
Num. obs.	26222	26222	26222	26222

Regressions weighted by posterior probability of a match between payroll data and voter file. The command information is only available for active officers (here as of October 2021). Level of observation: Employee. HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table A14: Correlation of Team Leadership and Team Composition

	Share of Party			Share of Race				
	Republican	Democrat	Non-Partisan	Hispanic	White	Asian	Black	Other
Republican leader	0.05*	-0.04*	-0.01	-0.03	0.02	-0.01	0.01	0.01
	(0.02)	(0.02)	(0.01)	(0.03)	(0.03)	(0.01)	(0.02)	(0.01)
Non-Partisan leader	-0.02	-0.03	0.05**	-0.01	0.01	-0.01	0.00	-0.00
	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.01)	(0.02)	(0.01)
Asian leader	0.04	-0.06	0.02	-0.02	-0.07	0.08***	-0.02	0.04*
	(0.04)	(0.04)	(0.03)	(0.04)	(0.04)	(0.01)	(0.02)	(0.02)
Black leader	-0.02	0.04	-0.02	-0.01	-0.13***	0.03*	0.08***	0.03**
	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.01)	(0.02)	(0.01)
Hispanic leader	-0.03	0.02	0.00	0.06**	-0.09***	0.01	0.01	0.01
	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.01)	(0.01)	(0.01)
Other Race leader	-0.03	0.05	-0.02	0.02	-0.10	0.02	-0.01	0.07***
	(0.05)	(0.05)	(0.03)	(0.04)	(0.05)	(0.02)	(0.03)	(0.02)
Mean DV	0.38	0.39	0.23	0.27	0.52	0.05	0.11	0.05
R ²	0.05	0.03	0.03	0.06	0.11	0.10	0.05	0.05
Adj. R ²	0.03	0.01	0.01	0.04	0.09	0.08	0.03	0.03
Num. obs.	644	644	644	644	644	644	644	644

Cross-sectional OLS; The information on team assignment is only available for active officers (here as of October 2021). Level of observation: Team. Outcome: Share of relevant demographic per team. All regressions include fixed effects for team-size bin (5 categories), precinct team, special operations team, and leader rank; these absorbed terms are not reported. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table A15: Correlation of Team Diversity and Team Leadership

	Gini Coefficient	
	Party	Race
Republican	0.03 (0.02)	0.03 (0.02)
Non-Partisan	-0.01 (0.02)	0.03 (0.02)
Asian	-0.03 (0.04)	-0.11*** (0.03)
Black	0.06* (0.03)	-0.05* (0.02)
Hispanic	0.03 (0.02)	-0.01 (0.02)
Other Race	0.01 (0.05)	-0.10** (0.03)
Adj. R ²	0.13	0.22
Num. obs.	644	644

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. All regressions include fixed effects for team-size bin (5 categories), precinct team, special operations team, and leader rank; these absorbed terms are not reported. Level of observation: Team. Dependent variable: Gini index.

B.3 Exits

Table A16: Differences in Years on Force at Retirement, by Characteristics and Rank

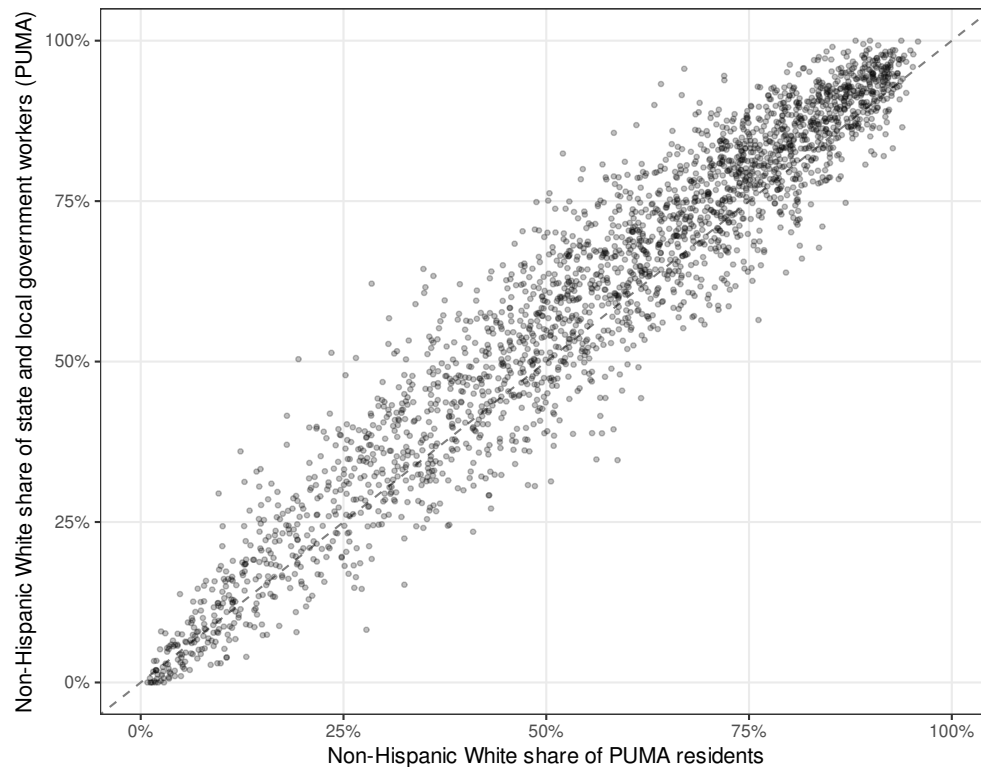
	Model 1	Model 2	Model 3	Model 4
Republican	0.52*** (0.13)		-0.04 (0.15)	-0.06 (0.15)
White		1.48*** (0.13)	1.50*** (0.14)	1.48*** (0.14)
Age at appointment	-0.12*** (0.01)	-0.11*** (0.01)	-0.11*** (0.01)	-0.11*** (0.01)
CCRB complaints				0.08*** (0.01)
Mean of DV	22.41	22.40	22.40	22.40
Adj. R ²	0.05	0.07	0.07	0.07
Num. obs.	6624	6613	6613	6613

OLS, weighted by probability of matches between payroll and voter file and payroll and retirement records. Level of observation: Retiree. Outcome: Time since appointment date at retirement. HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

C Additional Figures

C.1 Degree of Representativeness

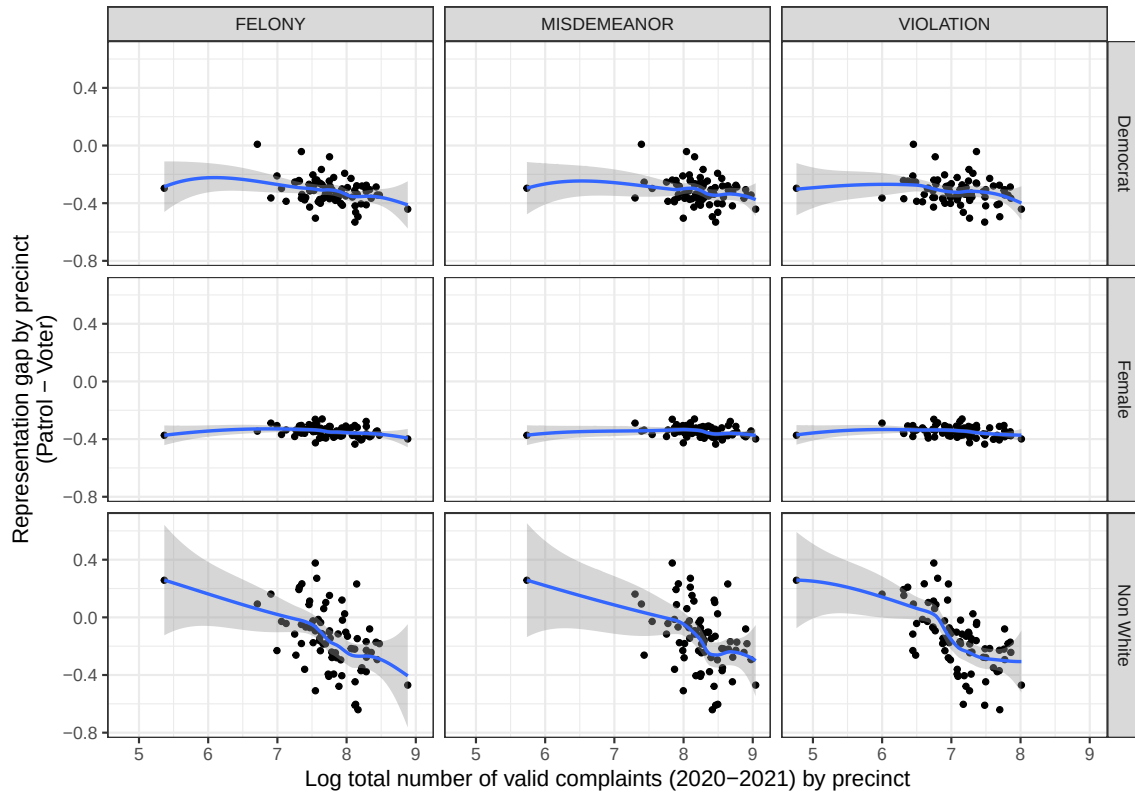
Figure A6: PUMA-level (Un)representativeness of State and Local Government Workforces



Note: Each point is one of 2,462 Public Use Microdata Areas (PUMAs) in the 2023 1-year ACS, units of approximately 100,000 residents. The vertical axis is the non-Hispanic White share of state and local government workers living in the PUMA; the horizontal axis is the non-Hispanic White share of all PUMA residents. Points above the 45-degree line indicate PUMAs in which the public workforce is Whiter than the resident population. Source: 2023 ACS 1-year Public Use Microdata Sample. The median PUMA-level gap is +3.9 percentage points.

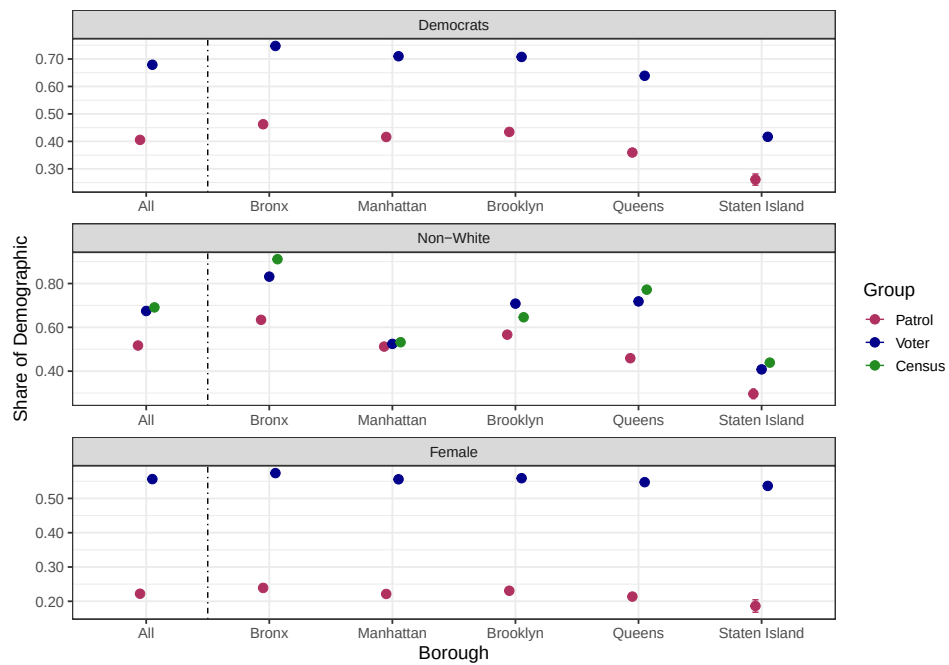
C.2 Entry

Figure A7: Correlation between Precinct Crime Rates and Representation Gaps



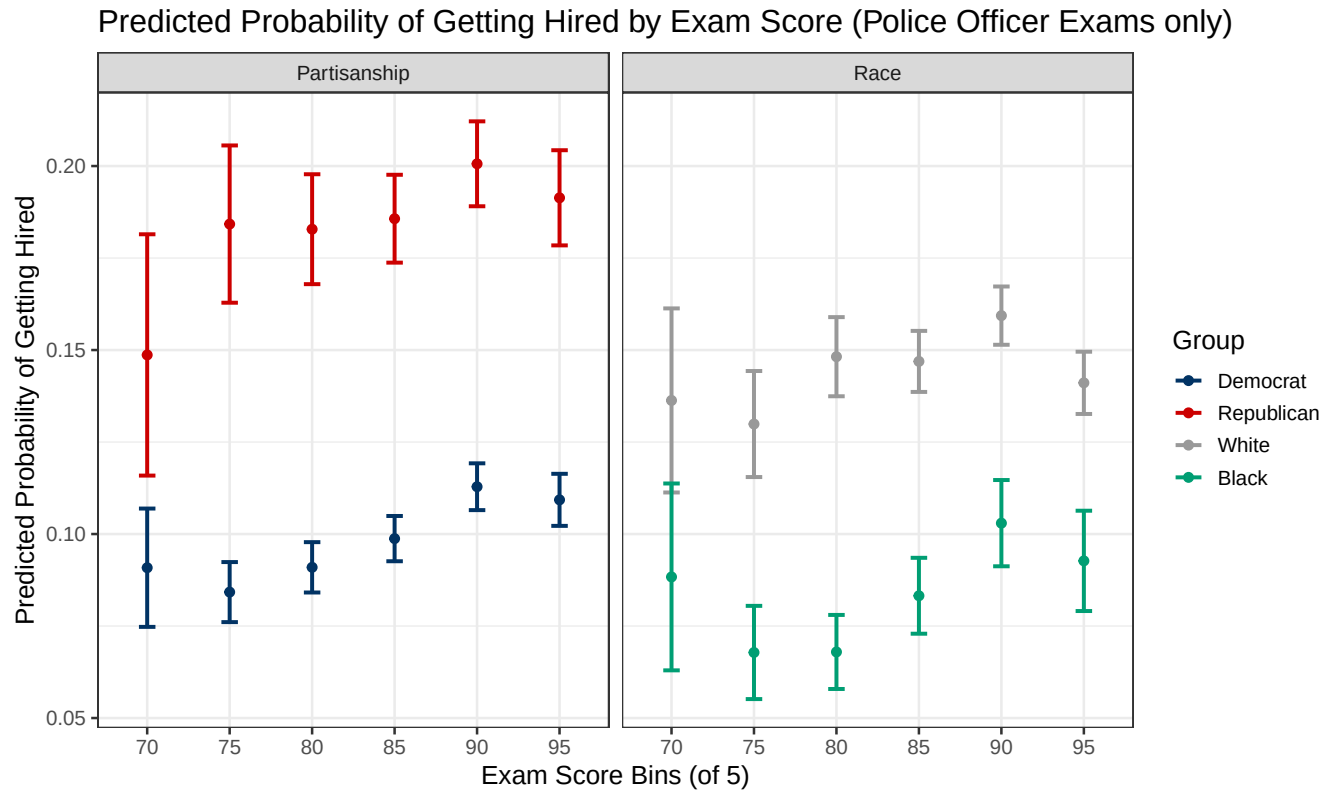
Note: The x-axis shows the log total number of valid complaints by precinct between 01/01/2020 and 30/06/2021. The y-axis is the representation gap by precinct (share of voter - share of officers), where lower values indicate an underrepresentation of the characteristics. The officer data only includes officers that are identifiable in my snapshot of the NYPD online profiles (see Appendix A.4).

Figure A8: Share of Demographics, Police vs. Registered Voters in NYC



Note: Uniformed police employees and traffic enforcement agents are assigned to boroughs based on their work location in their final year on the payroll. Agency estimates are weighted by the posterior probability of matches between agency payrolls and the voter file.

Figure A9: Hiring Differences Across Exam Scores



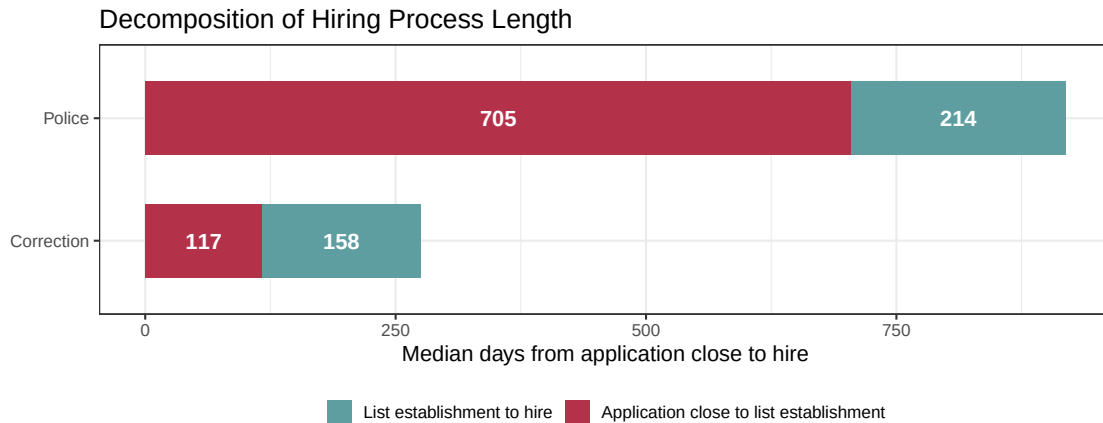
Note: Predicted probability of being hired by exam-score bin and group, from separate regressions interacting score bin with race and party. All regressions include exam fixed effects. These are held at their observed values for predictions.

Figure A10: Score Differences by Hiring Groups



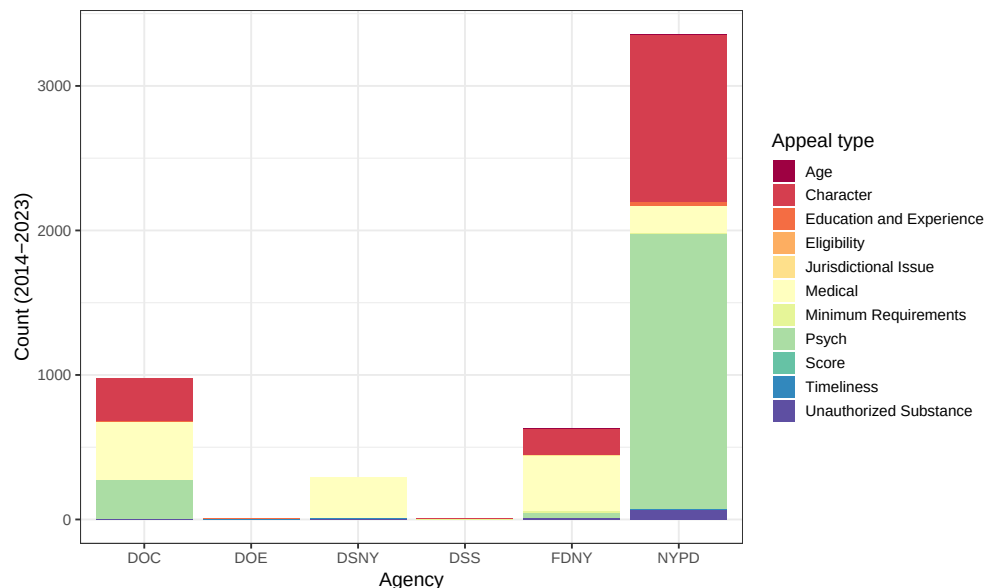
Note: Predicted exam scores by group and hiring status, from regressions of adjusted exam score on group interacted with hire status. All regressions include exam fixed effects. These are held at their observed values for predictions.

Figure A11: Length of Hiring Process



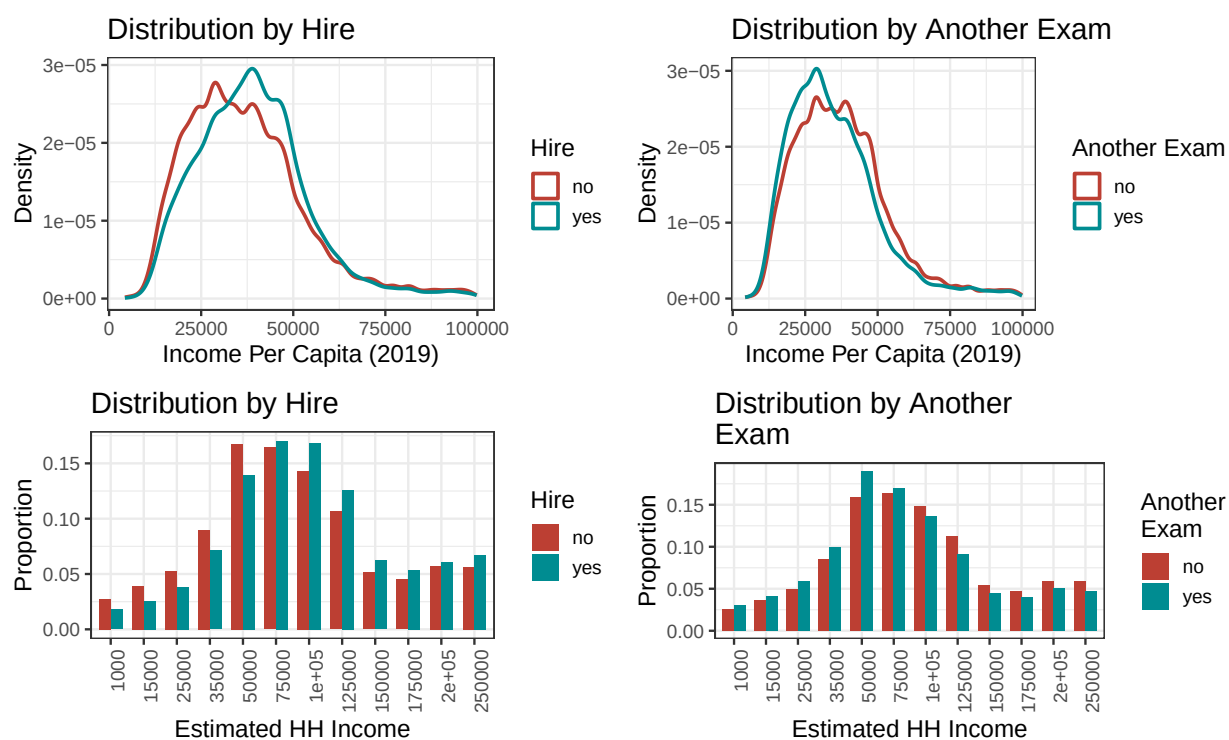
Note: Median number of days, decomposed into two stages. Application close to list establishment captures the period before NYPD/DOC requests certification of an eligible list from DCAS; under NYC civil service rules, certifications are initiated by the appointing agency to fill specific vacancies, so this stage mostly reflects each agency's pace of vacancy-driven requests rather than DCAS processing time. List establishment to hire captures the post-certification screening period administered by the agency itself (psychological, medical, character investigation, etc.). Sample restricted to candidates with non-missing application-period dates from the FOIA-released DCAS schedules and the DCAS open-data Annual Examination Schedule (police: 7,349 hires; corrections: 4,855 appointments).

Figure A12: Disqualification Appeals by Agency



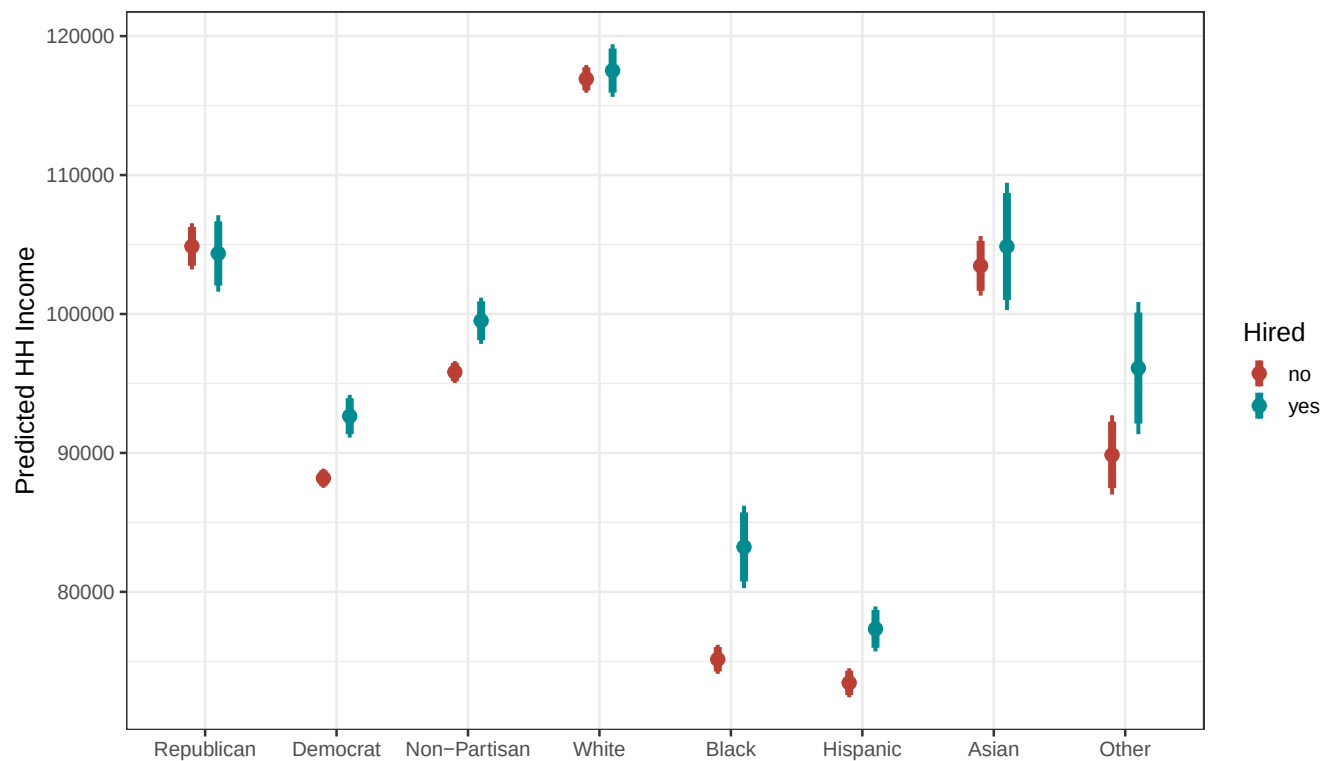
Note: Counts of disqualification appeals filed with the NYC Civil Service Commission, by agency.

Figure A13: Income Distributions by Exam Passer Type



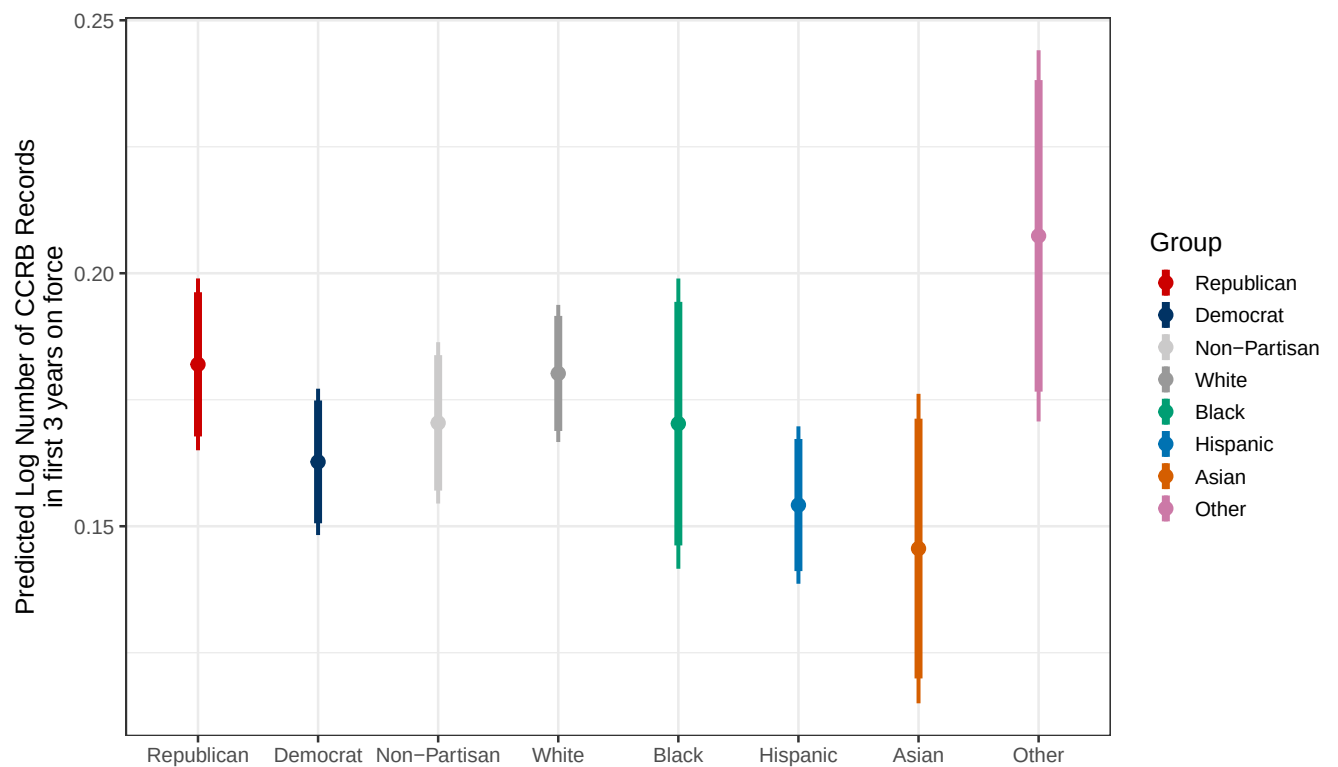
Note: The top panels use income per capita from the 2019 American Community Survey (matched by census tract), and the bottom panels use L2's estimated household income.

Figure A14: Predicted Household Income by Hiring Status and Groups



Note: Predicted household income from regressions of log income on hiring status interacted with the focal group; the cross-demographic (party in the race panels, race in the party panels) is included as a covariate. All regressions include exam-score-bin and exam-list fixed effects.

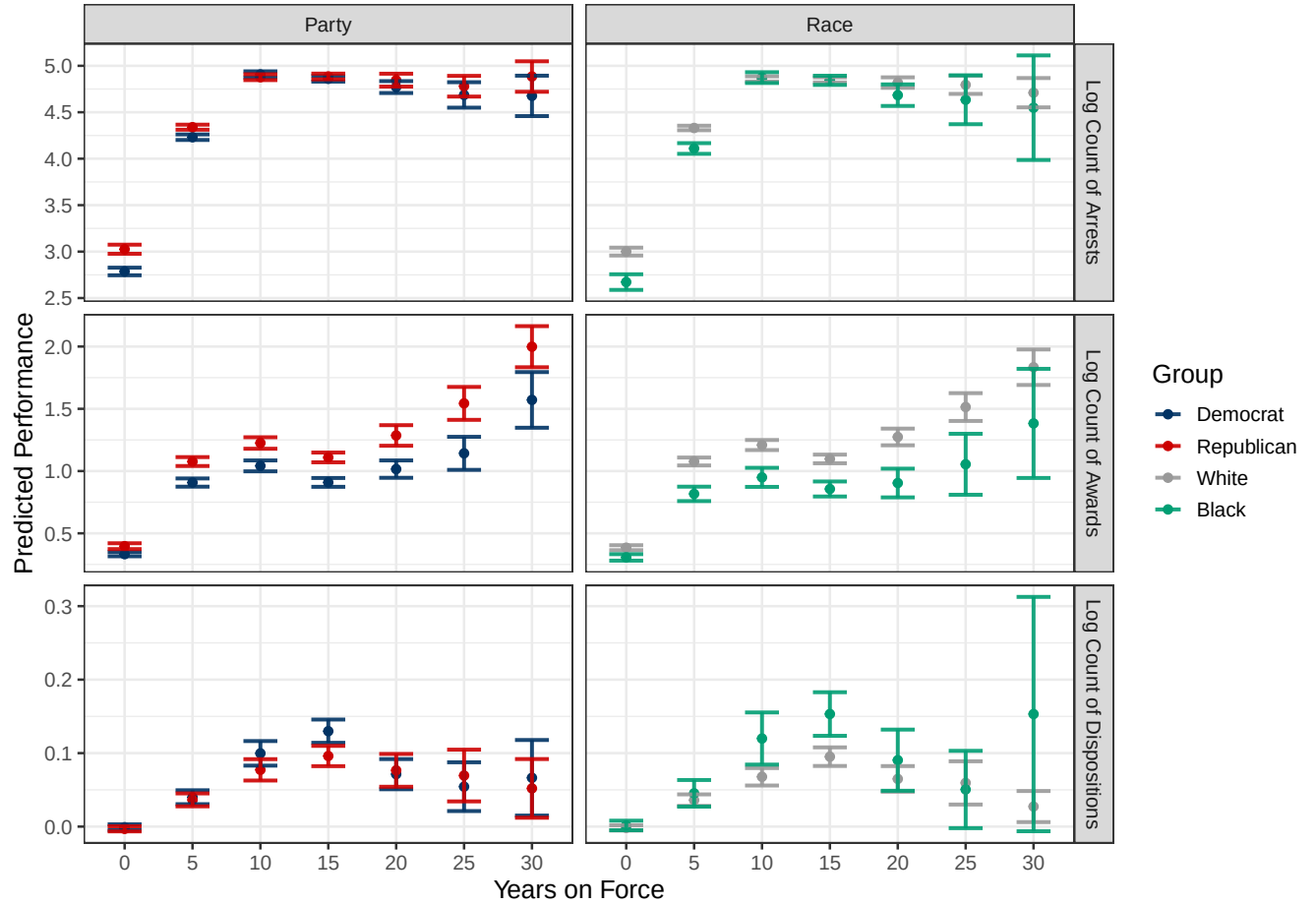
Figure A15: CCRB Records among Hired Police Officers 3 Years After Hiring



Note: Predicted log number of CCRB complaints in the first 3 years on the force, from a regression on race, party, and appointment-year fixed effects. Predictions integrate over the cross-demographic (party shown holding race fixed at observed values, and vice versa).

C.3 Career Progression

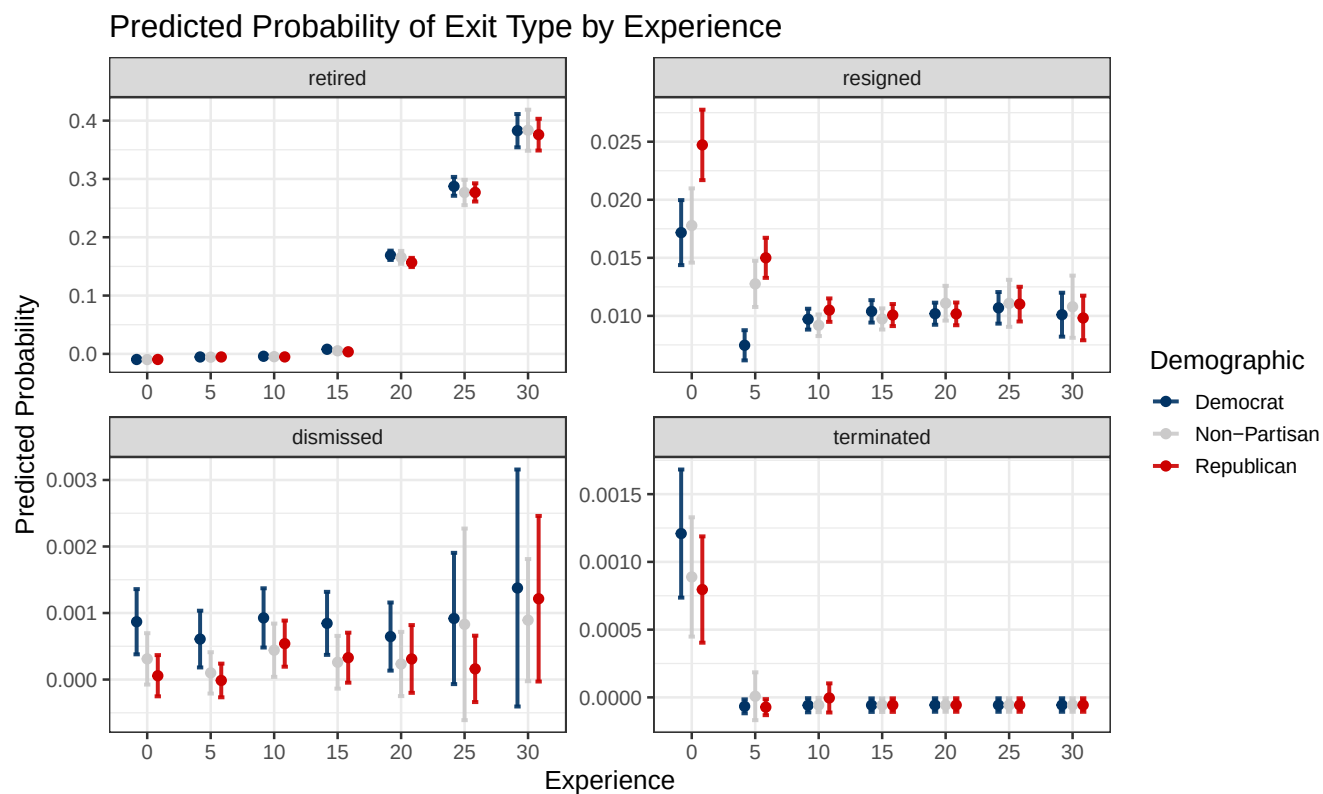
Figure A16: Difference in Behavior on the Job by Demographic and Tenure



Note: Predicted differences in on-the-job outcomes from regressions of each outcome on demographic interacted with tenure bins, with officer rank fixed effects. These are held at observed values for predictions.

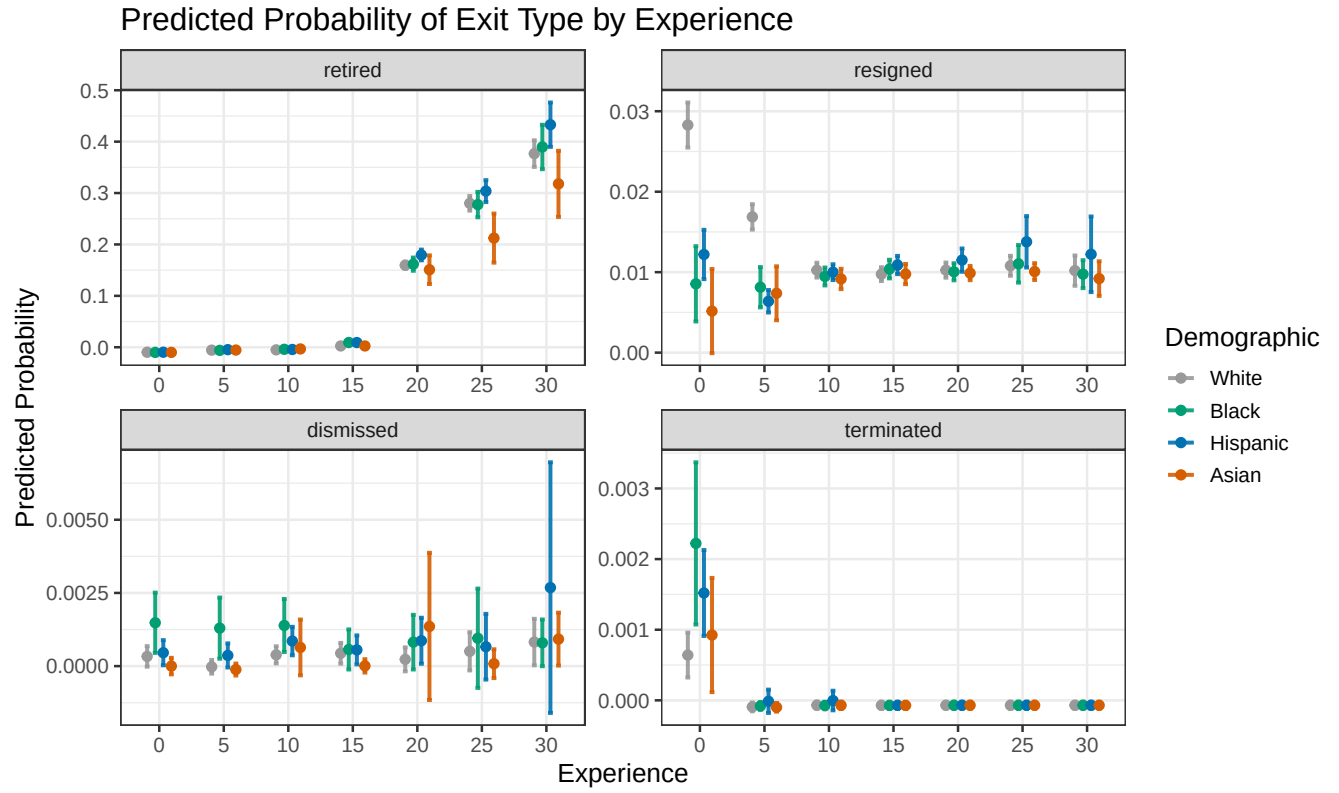
C.4 Exits

Figure A17: Predicted Probability of Exit Type by Years of Experience – Party



Note: The estimates are obtained from regressions of exit type (conditional on exit) on party interacted with experience bins, with appointment-year fixed effects. These are fixed at their observed values for predictions.

Figure A18: Predicted Probability of Exit Type by Years of Experience – Race



Note: The estimates are obtained from regressions of exit type (conditional on exit) on race interacted with experience bins, with appointment-year fixed effects. These are fixed at their observed values for predictions.

D Full Model Results

Table A17: Table 1 with Full Covariates

	Model 1	Model 2	Model 3	Model 4
	(0.00)	(0.00)	(0.00)	(0.00)
Republican	0.09***		0.08***	0.07***
	(0.00)		(0.00)	(0.00)
Non-Partisan	0.04***		0.03***	0.03***
	(0.00)		(0.00)	(0.00)
Black		-0.06***	-0.03***	-0.01***
		(0.00)	(0.00)	(0.00)
Hispanic		-0.02***	0.00	0.01**
		(0.00)	(0.00)	(0.00)
Asian		-0.00	0.02**	0.02**
		(0.01)	(0.01)	(0.01)
Other Race		-0.01**	0.00	0.01
		(0.00)	(0.01)	(0.01)
Exam Score Q2 (within exam)	0.01**	0.01*	0.01*	0.01*
	(0.00)	(0.00)	(0.00)	(0.00)
Exam Score Q3 (within exam)	0.02***	0.02***	0.02***	0.02***
	(0.00)	(0.00)	(0.00)	(0.00)
Exam Score Q4 (within exam)	0.03***	0.03***	0.02***	0.03***
	(0.00)	(0.00)	(0.00)	(0.00)
Exam Score Q5 / top (within exam)	0.02***	0.02***	0.02***	0.02***
Male				0.02***
				(0.00)
County: Kings				-0.01
				(0.00)
County: Nassau				0.03***
				(0.01)
County: New York				-0.01
				(0.01)
County: Queens				0.01
				(0.00)
County: Richmond				0.01
				(0.01)
County: Rockland				0.04***
				(0.01)
County: Suffolk				0.01*
				(0.01)
County: Westchester				-0.01
				(0.01)
Age				-0.00***
				(0.00)
NYPD Exam Cohort FE	Yes	Yes	Yes	Yes
Mean of DV	0.13	0.13	0.13	0.13
Adj. R ²	0.04	0.03	0.04	0.04
Num. obs.	75134	72851	69406	64340

Linear probability regressions, weighted by the posterior probability of a match between exam data and voter file, and between exam data and appointments. Reference categories: Democrat (party), White (race), Female (gender), and Bronx (county). HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A18: Table 2 with Full Covariates

	Model 1	Model 2	Model 3	Model 4
Republican	−0.010* (0.004)	0.001 (0.004)	−0.005** (0.001)	−0.004** (0.002)
Non-Partisan	−0.014*** (0.003)	−0.006 (0.004)	−0.004** (0.001)	−0.003* (0.001)
Black	0.084*** (0.005)	0.063*** (0.005)	0.016*** (0.002)	0.016*** (0.002)
Hispanic	0.032*** (0.003)	0.022*** (0.004)	0.012*** (0.001)	0.012*** (0.002)
Asian	0.039*** (0.006)	0.024** (0.008)	0.002 (0.002)	0.002 (0.002)
Other Race	0.049*** (0.006)	0.021** (0.007)	0.005** (0.002)	0.004 (0.002)
Male		0.029*** (0.003)		0.005*** (0.001)
County: Kings		0.014* (0.006)		−0.001 (0.002)
County: Nassau		−0.039*** (0.006)		−0.003 (0.002)
County: New York		−0.017* (0.007)		−0.001 (0.003)
County: Queens		0.004 (0.006)		0.001 (0.002)
County: Richmond		0.030*** (0.008)		0.000 (0.003)
County: Rockland		−0.067*** (0.009)		−0.009** (0.003)
County: Suffolk		−0.065*** (0.006)		−0.002 (0.002)
County: Westchester		−0.038*** (0.007)		0.001 (0.003)
Age		0.002*** (0.000)		0.000 (0.000)
NYPD Exam×Score Bin FE	Yes	Yes	Yes	Yes
Mean of DV	0.149	0.147	0.019	0.019
Adj. R ²	0.035	0.043	0.020	0.019
Num. obs.	62068	57389	60249	55759

Linear probability regressions, weighted by the posterior probability of a match between the respective data sets. Dependent Variables: Taking another civil service exam within a year of NYPD exam (Models 1, 2); Being appointed as corrections officer (Models 3, 4). Reference categories: Democrat (party), White (race), Female (gender), and Bronx (county). HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A19: Figure 4 – Full Coefficients (Party-Interacted Models)

	(1) Appoint.	(2) Officers	(3) Name	(4) Arrests
Appointed Prev. Year by Tract	0.005* (0.002)			
Officers by Tract (log)		0.003* (0.002)		
Arrests by Tract (log)				−0.001 (0.001)
Republican	0.017*** (0.003)	0.000 (0.010)	0.014** (0.004)	0.019 (0.011)
Non-Partisan	0.008*** (0.002)	0.004 (0.008)	0.012** (0.004)	0.004 (0.008)
Appoint. x Republican	0.003 (0.005)			
Appoint. x Non-Partisan	0.001 (0.004)			
Officers x Republican		0.006 (0.003)		
Officers x Non-Partisan		0.002 (0.003)		
Arrests x Republican				−0.001 (0.003)
Arrests x Non-Partisan				0.000 (0.002)
Population (tract)	−0.000 (0.000)	−0.000 (0.000)		0.000 (0.000)
Name Commonality			−0.000 (0.000)	
Exam FE	Yes	Yes	Yes	Yes
Score Bin FE	Yes	Yes	Yes	Yes
Num. obs.	59365	66850	58958	38838
Adj. R ² (full model)	0.036	0.032	0.032	0.022

Linear probability regressions, weighted by posterior probability of a match between exam data and voter file. Outcome: hired at NYPD. Reference category: Democrat. HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A20: Figure 4 – Full Coefficients (Race-Interacted Models)

	(1) Appoint.	(2) Officers	(3) Name	(4) Arrests
Appointed Prev. Year by Tract	0.006* (0.003)			
Officers by Tract (log)		0.010*** (0.002)		
Arrests by Tract (log)				−0.002 (0.002)
Black	−0.014*** (0.003)	−0.003 (0.010)	−0.010* (0.004)	−0.011 (0.010)
Hispanic	−0.001 (0.003)	0.023* (0.009)	−0.002 (0.005)	0.008 (0.009)
Asian	−0.003 (0.005)	0.014 (0.014)	0.001 (0.007)	−0.019 (0.015)
Other Race	0.003 (0.004)	0.009 (0.013)	0.005 (0.005)	−0.009 (0.014)
Appoint. x Black	0.003 (0.005)			
Appoint. x Hispanic	−0.004 (0.004)			
Appoint. x Asian	−0.000 (0.007)			
Appoint. x Other	−0.010 (0.007)			
Officers x Black		−0.004 (0.003)		
Officers x Hispanic		−0.008** (0.003)		
Officers x Asian		−0.005 (0.005)		
Officers x Other		−0.002 (0.005)		
Arrests x Black				0.000 (0.002)
Arrests x Hispanic				−0.001 (0.002)
Arrests x Asian				0.004 (0.004)
Arrests x Other				0.005 (0.004)
Population (tract)	−0.000 (0.000)	−0.000 (0.000)		−0.000 (0.000)
Name Commonality			−0.000 (0.000)	
Exam FE	Yes	Yes	Yes	Yes
Score Bin FE	Yes	Yes	Yes	Yes
Num. obs.	57551	64865	59093	37782
Adj. R ² (full model)	0.034	0.031	0.030	0.023
Adj. R ² (proj model)				

Linear probability regressions, weighted by posterior probability of a match between exam data and voter file. Outcome: hired at NYPD. Reference category: White. HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A21: Table 3 with Full Covariates

	Official Promotions			Title Changes		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Republican	−0.00 (0.00)	−0.00 (0.00)	−0.00 (0.00)	0.01* (0.00)	0.01 (0.00)	0.00 (0.00)
Non-Partisan	−0.00 (0.00)	−0.00 (0.00)	−0.00 (0.00)	−0.00 (0.00)	−0.00 (0.00)	−0.00 (0.00)
Black	−0.01 (0.00)	−0.01 (0.00)	−0.01 (0.00)	−0.03*** (0.01)	−0.02*** (0.01)	−0.03*** (0.01)
Hispanic	−0.01** (0.00)	−0.01** (0.00)	−0.01** (0.00)	−0.02*** (0.00)	−0.02*** (0.00)	−0.02*** (0.00)
Asian	0.04*** (0.01)	0.03*** (0.01)	0.04*** (0.01)	−0.00 (0.01)	−0.01 (0.01)	−0.00 (0.01)
Other Race	0.03*** (0.01)	0.03*** (0.01)	0.03*** (0.01)	−0.00 (0.01)	−0.01 (0.01)	−0.01 (0.01)
Age at Appointment		−0.00** (0.00)	−0.00** (0.00)		−0.00*** (0.00)	−0.00*** (0.00)
Male		0.08*** (0.01)	0.07*** (0.01)		0.17*** (0.02)	0.15*** (0.02)
Total CCRB Complaints			0.00*** (0.00)			0.01*** (0.00)
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Exposure FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of DV	0.09	0.09	0.09	0.17	0.17	0.17
Adj. R ²	0.10	0.10	0.11	0.19	0.19	0.20
Num. obs.	45990	43573	43573	45990	43573	43573

Linear probability regressions, weighted by the posterior probability of a payroll and voter file match. Reference categories: Democrat (party), White (race), and Female (gender). HC1 standard errors in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.